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Architecting a Multihomed Secondary Internet Network for Enhanced Reliability

Design, Implementation, and Performance Evaluation

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Abstract

This thesis investigates the design, implementation, and evaluation of a dual-ISP multihomed network architecture based on a customer-approved Low-Level Design (LLD). The work focuses on a lab-validated proof-of-concept rather than a full production deployment. A functional implementation was built in a Cisco IOS-based homelab, preserving the routing behavior and service separation objectives of the original design while omitting production-grade redundancy features such as firewall clustering, VSS/StackWise, and BFD.

The proof-of-concept incorporates VLAN segmentation, internal BGP (iBGP), Policy-Based Routing (PBR), Network Address Translation (NAT), and automatic ISP failover using IP SLA tracking. Controlled experiments were conducted to validate end-to-end connectivity, traffic steering behavior, failover performance, and baseline network metrics under lab conditions. Dedicated and shared services were correctly routed via their intended ISP during normal operation, and automatic failover successfully redirected traffic to the backup ISP during simulated outages. The measured failover time was approximately four seconds with 0.4% packet loss, and performance measurements showed stable low-latency connectivity and negligible packet loss and jitter within the homelab environment.

The findings demonstrate that the adapted homelab implementation successfully validates the core multihoming objectives of the original LLD under controlled conditions. The thesis also discusses the trade-offs introduced by replacing production technologies with Cisco IOS mechanisms and provides recommendations and future work for extending the proof-of-concept toward a production-ready multihomed deployment.

Keywords: Multihoming, Policy-Based Routing, iBGP, IP SLA, BFD, NAT, ISP Failover, Cisco IOS, Enterprise Network Design

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Terminology

Multihoming Connecting a network to two or more upstream Internet Service Providers (ISPs) to improve redundancy, availability, and traffic engineering flexibility.

Policy-Based Routing (PBR) A routing technique that forwards packets according to administrator-defined policies such as source IP address, VLAN, or application type, rather than relying solely on the destination routing table.

BGP Border Gateway Protocol. The standard inter-domain routing protocol used for exchanging routing information between autonomous systems. In this thesis, internal BGP (iBGP) is used between core and distribution devices.

IP SLA Cisco IOS functionality used to measure network performance and detect link failures through active network probes. In this project, IP SLA is used to trigger ISP failover events.

NAT Network Address Translation. A mechanism used to translate private IP addresses into public IP addresses for outbound network communication.

Failover The automatic redirection of network traffic to a backup ISP or alternative path when the primary connection becomes unavailable.

VLAN Virtual Local Area Network. A logical network segmentation technique used to isolate dedicated and shared services into separate broadcast domains.

SVI Switched Virtual Interface. A logical Layer 3 interface configured on a switch to provide inter-VLAN routing functionality.

iBGP Internal Border Gateway Protocol. A BGP session established between routers operating within the same autonomous system.

NTP Network Time Protocol. A protocol used for clock synchronization between network devices, required for accurate one-way latency measurements in IP SLA jitter operations.

PAT Port Address Translation. A form of NAT that allows multiple internal hosts to share a single public IP address using different transport-layer port numbers.

EtherChannel A link aggregation technology that combines multiple physical interfaces into a single logical interface to provide redundancy and increased bandwidth.

HSRP Hot Standby Router Protocol. A first-hop redundancy protocol that provides a virtual default gateway for end devices.

BFD Bidirectional Forwarding Detection. A rapid failure detection protocol capable of sub-second link monitoring and fast routing convergence.

High-Level Design (HLD) A conceptual architectural description that outlines the overall structure of a network, including major components, logical relationships, and design principles. The HLD defines what the system should achieve without specifying detailed configurations.

Low-Level Design (LLD) A detailed technical specification that describes exact configurations, protocols, addressing plans, routing behavior, and device-level implementation details. The LLD translates the HLD into deployable, device-specific instructions.

Proof-of-Concept (PoC) A limited-scope implementation used to validate key design principles, routing logic, and failover behavior under controlled conditions. A PoC demonstrates feasibility but does not include full production-grade redundancy or scale.

Link Aggregation Control Protocol (LACP) A protocol defined in IEEE 802.1AX that allows multiple physical Ethernet links to be combined into a single logical link (EtherChannel). LACP provides increased bandwidth and redundancy by dynamically negotiating and maintaining aggregated links.

Multipath TCP (MPTCP) A transport-layer protocol that allows a single TCP connection to use multiple network paths simultaneously. Relevant in multihoming contexts but not implemented in this thesis.

Software-Defined Wide Area Network (SD-WAN) A WAN architecture that uses centralized control and overlay tunnels to optimize traffic across multiple WAN links. Mentioned as a modern alternative to traditional multihoming.

Service Level Agreement (SLA) A contractual agreement between a service provider and a customer defining expected service metrics such as availability, latency, packet loss, and response times. This term should not be confused with Cisco IP SLA, which is a network monitoring feature used in this thesis.

Firewall Cluster A high-availability firewall deployment consisting of two or more nodes sharing state, configuration, and a virtual IP. Provides redundancy through active/standby or active/active modes. (Production design uses this; PoC does not implement it.)

ClusterXL Check Point's proprietary firewall clustering technology supporting load sharing, state synchronization, and failover. ClusterXL forms part of the original production design but was not implemented or validated in the proof-of-concept.

Check Point Gaia The operating system used by Check Point firewalls, combining features from IPSO and SecurePlatform. Relevant to the production design but not used in the PoC.

1 Introduction

Modern organizations depend heavily on continuous Internet connectivity for customer services, internal operations, and remote work. A failure in the secondary Internet connection can lead to customer outages, an increased support workload, and operational downtime. In the client's current environment, the secondary Internet consists of several low-capacity DSL(Digital Subscriber Line) circuits, which provide limited redundancy and require manual intervention whenever new customer services are added. This increases operational complexity and introduces a risk of service disruption.

To address these challenges, the client plans to redesign the secondary Internet infrastructure at two sites. The target production design includes two 1 Gbps multihomed Internet connections per site, delivered by two independent Internet Service Providers (ISP A and ISP B), and a redundant firewall cluster at the network edge. The design supports two service categories: **dedicated services** using public IP addressing and **shared services** using private addressing with NAT. This separation enables predictable routing behavior and improved manageability.

The customer-approved High-Level Design (HLD) and Low-Level Design (LLD) also include a Virtual Switching System (VSS) for switch-level redundancy. However, due to time and hardware constraints, the proof-of-concept (PoC) implementation in this thesis does not include VSS, firewall clustering, or Bidirectional Forwarding Detection (BFD). Instead, the PoC focuses on validating the core routing logic, service separation, and failover behavior using Cisco IOS devices in a controlled homelab environment. The objective is not to reproduce the production environment in its entirety, but to validate the key routing, failover, and service-separation mechanisms defined in the customer-approved LLD.

1.1 VSS Consideration

The production design incorporates a Virtual Switching System (VSS) at the core, distribution, and access layers to provide chassis-level redundancy and simplified management. While VSS is not implemented in the PoC, this thesis:

- Describes how VSS integrates with the multihomed architecture.
- Highlights its benefits for redundancy and operational simplicity.
- Demonstrates compatibility with dual-ISP connectivity in design diagrams.

1.2 Background and Motivation

Enterprise networks increasingly rely on high availability, redundancy, and flexible traffic management. Multihoming that is, connecting to multiple ISPs is a widely adopted strategy to improve availability and performance [3]. Without structured design, however, multihomed networks may suffer from inefficient routing, unpredictable failover behavior, and increased operational complexity [4].

Alternative technologies such as Multipath TCP (MPTCP) and Software-Defined WAN (SD-WAN) offer advanced capabilities including application-aware routing and dynamic path selection [5,6]. However, these approaches differ significantly from traditional network-layer multihoming. MPTCP requires end-to-end host support, while SD-WAN depends on centralized controllers and vendor-specific implementations.

In contrast, network-layer multihoming using BGP and Policy-Based Routing (PBR) provides full administrative control over routing decisions without requiring changes to end systems. This makes it well-suited for enterprise environments where compatibility, scalability, and deterministic failover behavior are essential.

The client's existing secondary Internet design is single-homed and manually managed. Redesigning it into a structured multihomed architecture with service separation and automated failover will:

- Improve availability and resilience

- Simplify customer onboarding and service delivery
- Reduce operational risk and manual configuration effort

1.3 Overall Aim and Problem Statement

1.4 Aim

The aim of this project is to design, implement, and evaluate a multihomed secondary Internet network that:

- Provides redundancy through dual ISPs
- Delivers dedicated and shared services with clear separation
- Improves availability and scalability
- Enhances manageability through structured routing and traffic control

1.5 Problem Statement

This thesis investigates how to architect a secondary Internet network that:

- Enforces service-based access policies (public dedicated vs. private shared)
- Integrates with an existing multihomed environment
- Provides reliable failover performance under controlled failure scenarios

1.6 Measurable Goals

The project is divided into the following measurable goals:

1. Redesign the secondary Internet infrastructure using existing hardware.
2. Implement routing mechanisms (iBGP, static routing) and Policy-Based Routing.
3. Create separate dedicated (ISP A) and shared (ISP B) service instances using VLAN segmentation and service-based routing policies.
4. Validate NAT, VLAN segmentation, and service separation.
5. Evaluate failover performance using:
    - Failover time (seconds)
    - Packet loss during failover (%)
    - Latency (ms)
    - Jitter (ms variation)

1.7 Research Questions

1. How can an existing secondary Internet network be redesigned to provide high availability using dual ISPs and multihoming principles?
2. How does the proposed network perform in terms of failover time, packet loss, and service reliability?
3. How can service-based access policies be enforced effectively across multiple ISPs?

1.8 Scope

1.9 Included

- Design and proof-of-concept implementation of a multihomed secondary Internet network
- Implementation of routing, NAT, VLAN segmentation, and failover testing
- Service differentiation per ISP
- Performance evaluation under simulated failure conditions

1.10 Excluded

- Full VSS deployment
- Firewall clustering
- BFD-based sub-second failover
- Major redesign of unrelated core services
- High-Level Design (HLD) development

1.11 Methodology

All testing and configuration were carried out in a controlled homelab environment using real Cisco IOS hardware. Failure scenarios were simulated by intentionally disrupting ISP connectivity (e.g., interface shutdowns and IP SLA target withdrawal) to replicate outage conditions in a safe and repeatable manner.

1.12 Design

- Develop a detailed Low-Level Design (LLD)
- Analyze the existing topology and traffic requirements
- Model dual ISP connectivity and service separation
- Design PBR and iBGP routing behavior

1.13 Implementation

- Implement the design using physical homelab hardware
- Configure routing, NAT, VLANs, and service differentiation
- Perform controlled failover tests using IP SLA and route manipulation

1.14 Testing

- Simulate ISP failures to measure failover and convergence times
- Validate dedicated and shared service functionality
- Measure latency, packet loss, and jitter under normal and failure conditions
- Document results using tables and diagrams

1.15 Report Structure

The final thesis is structured as follows:

- **Chapter 1: Introduction** Presents the background, motivation, aim, problem statement, research questions, scope, and methodology overview.
- **Chapter 2: Theoretical Background** Describes relevant networking concepts such as multihoming, routing protocols, Policy-Based Routing (PBR), NAT, VLANs, failover mechanisms, and redundancy technologies. Clarifies the difference between production technologies (e.g., firewall clustering, VSS, BFD) and the mechanisms used in the proof-of-concept.
- **Chapter 3: Methodology** Explains the research approach, evaluation strategy, test design, measurement procedures, and reproducibility considerations for the proof-of-concept.
- **Chapter 4: Pre-Study** Evaluates alternative architectural approaches, analyzes the existing network environment, and motivates the selection of the final design.
- **Chapter 5: Design and Implementation** Presents the production Low-Level Design (LLD) and the adapted proof-of-concept implementation using real hardware. Clearly distinguishes between production design intent and homelab implementation.
- **Chapter 6: Results** Provides the outcomes of connectivity tests, PBR verification, failover measurements, and baseline performance metrics.
- **Chapter 7: Discussion** Interprets the results, discusses limitations, compares expected versus observed behavior, and evaluates the scientific and practical implications.
- **Chapter 8: Conclusion** Summarizes the findings, evaluates goal achievement, presents scientific contributions, and outlines limitations and future work.
- **Appendices** Provide equipment specifications, addressing plans, configuration listings, and supplementary material supporting reproducibility.

2 Theoretical Background

This chapter presents the theoretical foundation required to understand the design and implementation of the proposed network architecture. It introduces key concepts related to network communication, redundancy, routing, and service management. The chapter also reviews relevant scientific literature to provide context for the project and to highlight existing approaches in the field.

2.1 Network Communication and High Availability

Modern enterprise networks are required to provide high availability, reliability, and scalability to support business-critical services. Network downtime can result in significant operational and financial consequences, making resilient network design a fundamental requirement in network infrastructures.

High availability is typically achieved through redundancy, where multiple network paths, devices, or services are deployed to eliminate single points of failure. One widely adopted approach is multihoming, where a network connects to multiple Internet Service Providers to ensure continuous connectivity in the event of link or provider failure.

Studies have shown that multihomed networks significantly improve availability and performance compared to single-homed networks by reducing service disruption and improving fault tolerance [3, 7]. In addition, multihoming enables more efficient traffic distribution and load balancing across multiple links [8].

To evaluate these properties, network performance is typically measured using quantitative metrics such as failover time, packet loss, throughput, latency, and jitter. These metrics are used in this project to assess the effectiveness of the proposed network design under normal and failure conditions.

This project builds upon these principles by designing a network architecture that incorporates redundancy and service differentiation.

2.2 Core Network Technologies

2.2.1 Multihoming

Multihoming refers to the practice of connecting a network to multiple Internet Service Providers (ISPs) to improve reliability, availability, and performance. In a single-homed network, external connectivity depends entirely on one ISP, creating a single point of failure. If that ISP experiences an outage, the entire network loses Internet access.

By contrast, multihomed networks maintain connections to multiple ISPs, allowing traffic to be rerouted in the event of a failure. This improves fault tolerance and enables more flexible traffic management strategies, such as load balancing and policy-based routing.

As illustrated in Figure 1, multihoming is typically implemented using BGP or static routes to exchange reachability information. In this project, multihoming is implemented using two ISPs per site to ensure redundancy and continuous service availability.

A key consideration in this thesis is the use of documentation prefixes (RFC 5737) for all public IP examples. These ranges (e.g., 192.0.2.0/24, 198.51.100.0/24, 203.0.113.0/24) are reserved for documentation and prevent accidental overlap with real Internet resources.

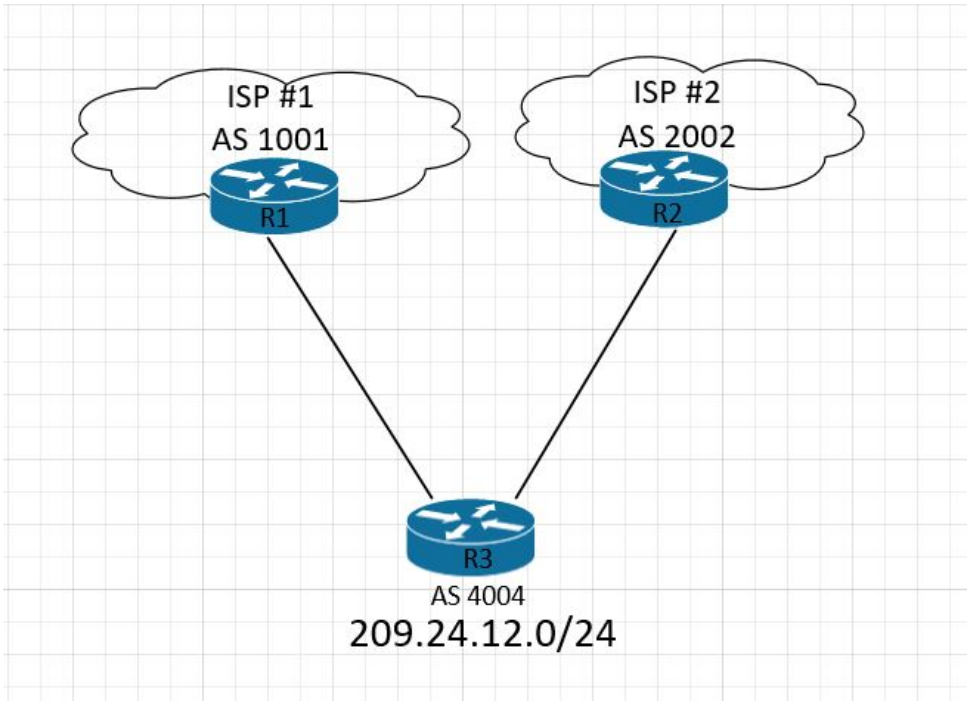


Figure 1: Overview of a multihomed network topology [1]

2.2.2 Border Gateway Protocol (BGP)

The Border Gateway Protocol (BGP) is the standard inter-domain routing protocol used on the Internet. It enables networks to exchange routing information and select optimal paths based on attributes such as AS-path length, local preference, and MED [9].

BGP can operate in two modes:

External BGP (eBGP): Used between different autonomous systems (e.g., enterprise to ISP).

Internal BGP (iBGP): Used within a single autonomous system to distribute externally learned routes.

In this project, iBGP is used internally between the core and distribution layers to ensure consistent route propagation. External connectivity is handled using static routes through the firewall or edge router, which simplifies the PoC and avoids the complexity of full eBGP peering with with upstream providers.

A well-known challenge with BGP is convergence delay, which occurs when the network recalculates routes after a failure. Convergence time directly affects packet loss and service continuity [10]. Studies show that convergence can range from seconds to minutes depending on topology and configuration [4].

The PoC design avoids external BGP convergence issues by using static routing toward the ISPs while still using iBGP internally to maintain routing consistency.

2.2.3 Policy-Based Routing (PBR)

Policy-Based Routing (PBR) allows routing decisions to be made based on administrator-defined policies rather than the destination IP address alone. Traditional routing selects paths solely based on the destination prefix. In contrast, PBR can match:

- Source IP

- Service type
- VLAN
- Application
- Interface

This makes PBR particularly useful in multihomed environments where different services must be routed through different ISPs.

In this project, PBR is used to enforce service separation:

- Dedicated services → routed via ISP-A
- Shared services → routed via ISP-B

This ensures predictable traffic behavior and aligns with the customer's operational model.

2.2.4 Network Address Translation (NAT)

Network Address Translation (NAT) maps private IP addresses to public IP addresses, enabling multiple internal hosts to share a limited pool of public addresses. NAT is widely used to conserve IPv4 space and provide a layer of abstraction between internal and external networks.

NAT is commonly used to conserve public IP address space and to provide a level of abstraction between internal and external networks. Different types of NAT exist, including static NAT for one-to-one mappings and dynamic or port address translation (PAT) for many-to-one mappings.

In the PoC, NAT is implemented on the edge router to support shared services, while dedicated services use direct public addressing.

2.2.5 Firewall and High Availability

Firewalls protect internal networks by filtering traffic based on security policies. In enterprise environments, firewalls are often deployed in high-availability (HA) configurations such as active-standby or active-active clusters. These designs ensure that if one firewall fails, another takes over seamlessly.

In the production design, the client uses a clustered firewall platform to provide redundancy and stateful failover.

However, in the proof-of-concept, a clustered firewall is not implemented due to hardware and licensing constraints. Instead, the PoC uses a Cisco IOS router to emulate the routing and failover behavior at the network edge. This approach preserves the architectural intent while remaining feasible within the homelab environment.

2.2.6 Virtual Switching System (VSS)

A Virtual Switching System (VSS) allows two physical switches to operate as a single logical switch. This provides:

- simplified management

- increased redundancy
- elimination of spanning-tree complexity
- multi-chassis EtherChannel support

Although VSS is part of the production design, it is not implemented in the PoC due to time and hardware limitations. Instead, the thesis:

- describes how VSS integrates with the multihomed architecture
- explains its redundancy benefits
- ensures the PoC remains compatible with a future VSS deployment

2.3 Related Work

Several studies have explored the impact of multihoming and routing mechanisms on network performance and reliability. In addition to presenting their findings, it is important to examine the methods, metrics, and conclusions of these studies.

Akella et al. [3] evaluate multihoming using a measurement-based approach on the real Internet. Their methodology involves monitoring path performance across multiple ISPs and comparing multihoming with overlay routing techniques. The study measures metrics such as latency, path availability, and failure recovery time. Their results show that multihoming can significantly reduce downtime by rapidly switching to alternative paths. They conclude that multihoming improves both reliability and end-to-end performance, particularly in unstable network conditions.

Shin et al. [7] use an analytical and simulation-based approach to study the impact of multihoming on routing stability. Their evaluation focuses on metrics such as packet loss, route stability, and network availability. The results indicate that while multihoming enhances fault tolerance, it can introduce routing instability if not properly managed. The authors conclude that careful routing policy design is necessary to fully benefit from multihoming.

Bennesby et al. [10] conducted a survey-based analysis of BGP convergence behavior across multiple studies. Their work focuses on convergence time, route propagation delay, and packet loss during routing updates. The study showed that BGP convergence can vary from seconds to several minutes, depending on network configuration and scale. They concluded that convergence delay is a critical factor affecting service availability in multihomed networks.

Basit et al. [4] performed a simulation-based experimental study to evaluate routing performance in multihomed environments. Their methodology involves controlled failure scenarios and traffic simulations. The study measures convergence time, throughput, latency, and packet loss. The results demonstrate that routing performance depends heavily on configuration and failover strategy. The authors concluded that optimized routing design is essential for achieving high availability and efficient failover.

These studies demonstrate that multihoming improves network availability and resilience, but also highlight the importance of routing convergence and traffic control. The performance metrics used in these works directly inform and justify the evaluation methodology adopted in this project. Consequently, the evaluation conducted in this thesis focuses on failover time, packet loss, latency, and jitter, allowing direct comparison

between the behaviour observed in the proof-of-concept implementation and the metrics commonly used in previous multihoming research.

To provide a clearer overview of how these studies differ in methodology and evaluation, their key characteristics are summarized in Table 1.

Study	**Method**	**Metrics**	**Key Findings**
Akella et al.	Real-world Internet measurements	Latency, availability, failover time	Multihoming significantly reduces downtime and improves path reliability
Shin et al.	Analytical + simulation	Packet loss, route stability	Improved availability but increased routing complexity
Bennesby et al.	Literature survey	Convergence time, packet loss	BGP convergence varies widely and impacts service availability
Basit et al.	Simulation-based experiments	Throughput, latency, packet loss	Performance depends heavily on routing configuration and failover strategy

Table 1: Summary of methods, metrics, and results in related work

While Table 1 presents a structured comparison of methods and metrics, a simplified qualitative comparison of their findings is provided in Table 2.

Study	**Availability**	**Failover Speed**	**Stability**	**Control**
Akella et al.	+++	++	++	+
Shin et al.	++	+	+	++
Bennesby et al.	++	+	++	++
Basit et al.	++	++	++	++

Table 2: Qualitative comparison of results from related work (+ low, ++ medium, +++ high performance)

Table 2 summarizes the findings of the reviewed studies using a qualitative scale derived from the authors' reported results and conclusions regarding availability, failover behaviour, routing stability, and traffic management capabilities. The comparison is intended as a high-level summary rather than a quantitative ranking.

Studies based on real-world measurements, such as Akella et al., demonstrate high availability but less control over network behavior. In contrast, simulation-based studies provide more stable and controlled results but may not capture all real-world dynamics.

This variation further highlights the importance of selecting an appropriate evaluation methodology, as discussed in this project.

2.4 Scientific Gap and Contribution

The reviewed studies demonstrate that multihoming improves availability and resilience, but they also reveal several limitations that motivate the need for this thesis. Existing research primarily evaluates multihoming using large-scale Internet measurements,

analytical models, or simulation-based environments. While these approaches provide valuable insights into routing behavior, convergence time, and failover performance, they do not examine how multihoming techniques can be adapted and validated within constrained enterprise environments or educational homelabs.

None of the reviewed studies investigate the combined use of Policy-Based Routing (PBR), IP SLA tracking, and service-based traffic separation within a practical enterprise-oriented proof-of-concept implementation using physical networking equipment. Similarly, previous work does not address how production-oriented designs such as clustered firewalls, VSS-based switching, or BFD-based sub-second failover can be translated into simplified laboratory environments while preserving core routing logic and failover behavior.

This thesis contributes to the field by:

- Demonstrating a practical proof-of-concept that validates multihoming, service separation, and failover behavior using real hardware in a controlled homelab environment.
- Evaluating failover performance using empirical measurements (latency, packet loss, and failover time) under repeatable failure scenarios.
- Highlighting the operational trade-offs between production-grade redundancy mechanisms and their simplified counterparts in a PoC environment.
- Providing a structured methodology that can be reproduced by practitioners, students, and organizations seeking to validate multihoming designs without full-scale production infrastructure.

By addressing these gaps, the thesis bridges the divide between theoretical multihoming research and practical implementation, offering insights into how enterprise-grade routing logic can be validated in resource-constrained environments.

3 Methodology

This chapter describes the methodology used to design, implement, and evaluate the proposed multihomed network architecture. The approach combines design science methodology with experimental evaluation to ensure that the results are measurable, reproducible, and aligned with the project objectives. Multihoming has been widely studied as a means to improve reliability and performance in IP networks [7, 8, 11].

The chapter is structured into the following sections: research approach, project method, and evaluation method. All implementation and testing are performed using real network hardware in a controlled lab environment, while failure conditions are simulated to evaluate network behavior.

3.1 Research Approach

A quantitative research approach is used, focusing on measurable network performance metrics such as failover time, packet loss, latency, and jitter [4, 10]. Quantitative evaluation is appropriate because the goal is to assess the behavior of routing and failover mechanisms under controlled conditions.

The methodology combines:

- **Design science methodology**, used to design and refine the network architecture.
- **Experimental research**, used to evaluate performance under controlled failure scenarios.

A controlled lab environment is used to ensure reproducibility. Although real hardware is used for implementation, failure scenarios are simulated through controlled physical disconnection of ISP-facing links, representing ISP outages in a safe and repeatable manner. This allows the same test to be executed multiple times under identical conditions.

The use of physical hardware, documented configurations, defined test procedures, and controlled failure scenarios enables other researchers or practitioners to reproduce the experiments and verify the results.

3.1.1 Research Deliverables

This project produces a network design artifact implemented as a proof-of-concept using real hardware. The artifact consists of:

- A detailed Low-Level Design (LLD) describing routing, addressing, and service separation.
- Routing, NAT, and PBR configurations adapted for the homelab environment.
- A controlled test environment enabling repeatable failover and performance measurements.

The outputs include measurable performance data such as failover time, packet loss, latency, and jitter. These outputs form the basis for the performance evaluation described in Section 3.3. Reproducibility is emphasized, consistent with prior research on routing convergence and multihoming behavior [5, 12].

3.2 Project Method Description

The project follows a design science methodology structured into a series of phases that guide the development and evaluation of the proposed network architecture. This approach is appropriate because the project involves the creation and validation of a technical artifact. Design Science Research (DSR) is appropriate because the project aims to create and evaluate a technical artifact, in this case, a multihomed network architecture and its proof-of-concept implementation rather than to study organizational processes.

3.2.1 Problem Identification

The initial phase involves analyzing the existing network infrastructure and identifying its limitations. The current use of single-homed DSL connections results in limited capacity, reduced availability, and increased operational complexity. These issues define the need for a more resilient and scalable solution.

3.2.2 Requirement Definition

Based on the identified problems, technical and operational requirements are defined. These include high availability through dual ISP connectivity, support for service differentiation (dedicated and shared services), predictable failover behavior, and centralized traffic control.

3.2.3 Design and Development

In this phase, a multihomed network architecture is designed to meet the defined requirements. The design phase produces a Low-Level Design that specifies routing behavior, service separation, and traffic engineering mechanisms. The design also considers production technologies such as firewall clustering and VSS, even though these are not implemented in the proof-of-concept.

3.2.4 Implementation

The implementation phase implements the designed architecture as a proof-of-concept using real hardware in a controlled lab environment. This enables full control over variables and repeatable testing. The implementation focuses on validating routing logic, PBR behavior, and failover mechanisms rather than replicating full production redundancy.

3.2.5 Testing and Measurement

The implemented network is subjected to controlled test scenarios to evaluate its performance. These include normal operations as well as simulated failure conditions such as ISP outages. Performance metrics, including failover time, packet loss, latency, and jitter, are collected for both dedicated and shared services. All tests were performed on identical hardware configurations to ensure consistent baseline performance. No other traffic was present in the lab environment, ensuring that measurements reflect only the behavior of the tested architecture. Each failover scenario was executed five times to ensure consistency and reduce the risk of anomalous results. Failover time was measured using continuous ICMP echo requests transmitted at one-second intervals from the test host. The underlying IP SLA tracking mechanism used a five-second probe frequency, as configured on the EDGE-FW router. This approach allowed failover convergence to be observed independently of the failure-detection mechanism.

3.3 Evaluation Method

The evaluation of the proposed network is based on quantitative performance measurements collected during controlled failure scenarios. The primary metrics include failover time, packet loss, latency, and jitter.

Failover time is measured as the interval between a simulated failure event (e.g., ISP link outage) and the restoration of connectivity. This is calculated using continuous ICMP probes at one-second intervals. Because probe frequency limits temporal resolution, the measurement uncertainty is approximately ± 1 second.

Packet loss is calculated as the percentage of lost packets during the transition period. Latency and jitter are measured using standard network testing tools within the lab environment.

Each test scenario is executed multiple times to ensure consistency. Results are compared across repetitions to identify variance and assess stability. The evaluation includes:

- Normal operation tests
- Failover tests for dedicated services
- Failover tests for shared services

These metrics were selected because they directly address the research questions concerning network availability, failover performance, service reliability, and the effectiveness of service-based traffic steering. The results are compared against the defined requirements to determine whether the proposed architecture meets the project objectives. The following chapter presents the technical implementation of the designed architecture, including configuration details and validation of the proposed solution.

4 Pre-study

This chapter presents the pre-study and system design phase of the project. It defines the technical requirements, evaluates alternative design approaches, and justifies the selection of the final architecture. The purpose of this phase is to ensure that the proposed solution aligns with the project objectives, including high availability, redundancy, service differentiation, and manageability.

4.1 Requirements

The following technical requirements were identified based on the client's existing network limitations and project objectives:

- High availability through redundant Internet connections
- Support for multihoming using at least two Internet Service Providers (ISPs)
- Separation of services into dedicated and shared services
- Fast failover in case of ISP or device failure
- Centralized and manageable traffic control
- Scalable architecture for future expansion
- Compatibility with the existing core network design

4.2 Approach Alternatives

Several architectural approaches were considered for implementing the proposed network.

4.2.1 Single ISP Architecture

A single ISP design relies on one upstream provider for Internet connectivity.

Advantages:

- Simple configuration
- Low cost

Disadvantages:

- Single point of failure
- No failover capability

4.2.2 Dual ISP with Static Routing

This approach uses two ISPs with static routing for traffic forwarding.

Advantages:

- Predictable behavior
- Easy to implement

Disadvantages:

- Slow failover due to manual or timer-based route withdrawal
- Limited scalability and flexibility

4.2.3 Dual ISP with BGP (Without Policy Control)

This approach uses external Border Gateway Protocol (eBGP) for dynamic routing between ISPs.

Advantages:

- Automatic failover
- High scalability

Disadvantages:

- Limited traffic control: routing decisions are primarily destination-based
- Service separation cannot be enforced without additional mechanisms

A key limitation is that BGP alone cannot guarantee that dedicated and shared services consistently use different ISPs. Without additional policy mechanisms, traffic may follow routes selected according to BGP path attributes rather than operational requirements. For example, if both ISPs advertise a default route, BGP may select ISP A for all outbound traffic based on path attributes, even though shared services are intended to use ISP B. This results in asymmetric routing and breaks the required service separation. By applying PBR, shared-service traffic is explicitly steered toward ISP B regardless of BGP's default path selection.

4.2.4 Advanced Multihoming with BGP and Policy-Based Routing (PBR)

This approach combines dynamic routing with policy-based traffic control.

Advantages:

- High availability
- Fine-grained traffic control
- Service-based routing aligned with operational requirements

Disadvantages:

- Increased implementation complexity

4.3 Comparison of Approaches

The alternatives were evaluated based on reliability, scalability, manageability, and implementation complexity. These criteria reflect the ability of each approach to meet the project requirements.

Approach	Reliability	Scalability	Manageability	Complexity	Total Score
Single ISP	-	-	+	+	0
Dual ISP + Static	0	0	0	0	0
Dual ISP + BGP	+	+	0	-	+1
BGP + PBR	+	+	+	-	+2

Table 3: Pugh matrix comparison of network design approaches

The comparison is performed using a Pugh matrix, where the Dual ISP with the static routing approach is used as the baseline. Each alternative is evaluated relative to the baseline using qualitative indicators: (+) indicates better performance, (0) indicates similar performance, and (–) indicates worse performance. In the case of complexity, a lower level of complexity is considered preferable and is therefore evaluated as a positive (+) outcome.

The scoring reflects the following reasoning:

- **Reliability:** Multihoming improves availability compared to single-homed designs.
- **Scalability:** BGP-based approaches scale better than static routing.
- **Manageability:** PBR enables service separation and centralized control.
- **Complexity:** BGP and PBR introduce configuration overhead compared to static routing.

The scoring is qualitative and intended to support design selection rather than provide a quantitative performance evaluation.

The results show that the BGP with PBR approach achieves the highest total score, indicating superior performance in terms of reliability, scalability, and traffic control. Although it introduces higher complexity, it best satisfies the project requirements and is therefore selected as the preferred solution.

For manageability, the PBR-based approach scores higher because it allows explicit service separation without requiring complex BGP policy tuning or prefix manipulation. In contrast, pure BGP solutions require more operational expertise to maintain route filters, communities, and path attributes. For complexity, BGP introduces additional configuration overhead compared to static routing, while PBR adds targeted complexity only where service-based control is required.

To further illustrate the structural differences between the evaluated approaches, Figure 2 presents common BGP connectivity models, ranging from single-homed to fully multihomed architectures.

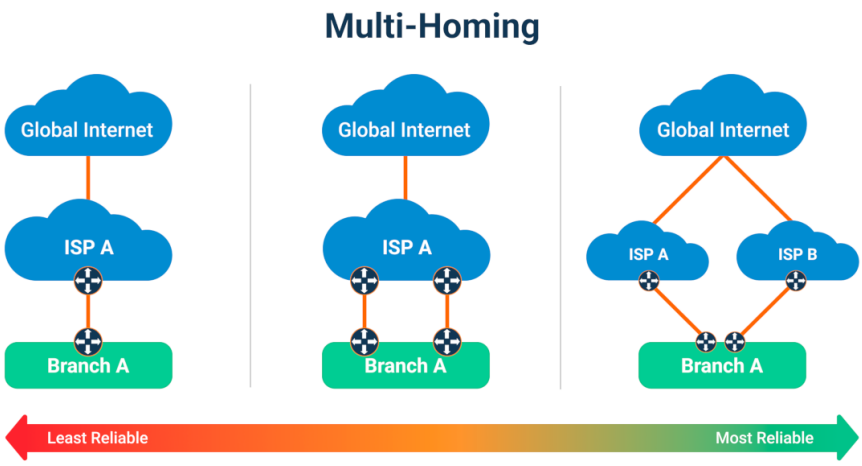


Figure 2: Overview of BGP connectivity models, including single-homed, dual-homed, and multihomed topologies [2]

As shown in Figure 2, single-homed designs rely on a single ISP and therefore introduce a clear single point of failure. Dual-homed architectures improve redundancy but still provide limited control over traffic distribution. Fully multihomed designs offer the highest level of availability and flexibility by enabling dynamic path selection across multiple ISPs.

It should be noted that while the models in Figure 2 illustrate traditional BGP-based multihoming using external BGP (eBGP), the proposed design adapts this concept by using internal BGP (iBGP) within the network and centralized policy-based routing at the firewall or edge router. This approach simplifies external routing while maintaining the benefits of multihoming.

4.4 Proposed Approach

Based on the requirements analysis and comparison of alternatives, the selected solution is a multihomed network architecture using dual ISPs, internal BGP (iBGP) for internal routing, static routing toward external ISPs, and policy-based routing (PBR) for service-based traffic control.

This approach provides:

- Reliable failover through redundant ISP connections
- Service-based traffic control using PBR

- Centralized and manageable routing behavior

Although this approach introduces additional complexity, it is justified by the significant improvements in reliability, scalability, and operational control. Furthermore, the architecture supports detailed low-level implementation, enabling network engineers to deploy and operate the solution effectively.

While the full production architecture includes additional redundancy features such as firewall clustering, VSS, and BFD, the implementation in this project is delivered as a proof-of-concept due to hardware and time constraints. The PoC preserves the functional behavior of the selected multihoming and policy-based routing design. It is important to note that PBR is not universally the best solution for multihoming; many enterprise networks rely solely on BGP for scalability and simplicity. However, for this specific requirement (strict separation between dedicated and shared services), PBR provides deterministic control that cannot be achieved with BGP alone without significantly increasing configuration complexity.

4.5 Rationale for Selecting Multihoming

Alternative technologies such as Multipath TCP (MPTCP) and Software-Defined WAN (SD-WAN) were considered.

MPTCP enables the use of multiple network paths at the transport layer but requires end-to-end support, limiting its applicability in heterogeneous enterprise environments. SD-WAN provides centralized control and dynamic path selection but introduces additional infrastructure requirements and vendor dependencies.

In contrast, multihoming with BGP:

- Is widely deployed in enterprise and service provider networks
- Provides full administrative control over routing decisions
- Supports controlled experimentation in a lab environment

This analysis justifies the selection of a multihomed architecture using BGP and policy-based routing. The detailed design and implementation of the proposed solution are presented in Chapter 5.

5 Design

This chapter presents the detailed design of the proposed multihomed network architecture and the corresponding proof-of-concept (PoC) implementation. To avoid confusion between the production Low-Level Design (LLD) and the simplified homelab environment, the chapter is structured into two parallel perspectives:

- **Production LLD:** The intended enterprise deployment, including clustered firewalls, VSS/StackWise switching, and redundant uplinks.
- **PoC Implementation:** A simplified but functionally equivalent routing environment implemented using Cisco IOS hardware.

This separation ensures that the routing logic, service separation, and failover behavior can be validated in the PoC without implying that full production-grade redundancy was implemented.

5.1 High-Level Design

The production architecture follows a hierarchical network model consisting of access, distribution, and core layers, integrated with dual Internet Service Providers and a clustered firewall at the network edge, as shown in Figure 3. The design supports multihoming, service separation, and high availability, consistent with established best practices in enterprise network design [13, 14].

However, due to hardware and time constraints, the PoC implementation does not include VSS/StackWise, firewall clustering, or multi-chassis redundancy. Instead, the PoC focuses on validating the routing logic, service separation, and failover mechanisms defined in the production LLD.

The following sections describe the production Low-Level Design, followed by the adapted PoC implementation.

5.2 Low-Level Design (LLD)

Introduction: The Low-Level Design specifies the detailed implementation of a multihomed network architecture using two 1 Gbps circuits provided by two ISPs. It defines the physical and logical topology, VLAN structure, IP addressing plan, link aggregation, routing behavior, and firewall policies. This LLD serves as an operational template for network engineers deploying the production environment.

The 1 Gbps ISP links are assumed to be best-effort broadband connections without formal SLA guarantees. This affects availability and support expectations, but not the core routing design.

All design diagrams and addressing tables use RFC 5737 documentation prefixes as defined in RFC 5737 [15]. However, the PoC lab used non-RFC public IPs (e.g., 209.165.1.0, 8.8.8.1, 9.9.9.1) during testing. These appear in configuration listings and screenshots to accurately reflect the real lab environment and do not represent real customer addresses.

5.3 Physical Topology

Production LLD: The production physical topology follows a hierarchical three-tier Cisco architecture:

- **Access Layer** — Layer 2 connectivity for customer edge devices and internal service networks.
- **Distribution Layer** — Aggregates access switches and provides Layer 3 termination for customer VLANs.
- **Core Layer** — High-speed routing backbone connecting the distribution layer to the firewall cluster.

Each layer consists of redundant Cisco Catalyst switches configured in StackWise or Virtual Switching System (VSS) mode. A pair of Check Point firewalls operates in an Active–Standby cluster and connects to both ISP Red and ISP Blue.

PoC Implementation: The PoC uses standalone Cisco IOS routers and switches without VSS, StackWise, or firewall clustering. The physical redundancy of the production design is not replicated; instead, the PoC preserves the logical routing behavior and failover mechanisms.

5.4 Logical Topology

Production LLD: The network is logically divided into two service domains:

- **Dedicated Services (ISP Red):** Public IP addressing, deterministic routing, no NAT. ISP Blue serves as failover.
- **Shared Services (ISP Blue):** Private addressing with NAT. ISP Red serves as failover.

Traffic steering is enforced using static routing and Policy-Based Routing (PBR) on the Check Point firewall [16].

PoC Implementation: The PoC uses Cisco IOS PBR to emulate the Check Point PBR logic. Static routes and IP SLA tracking are used to simulate ISP failover behavior.

5.5 Addressing Plan

A maximum of 10 dedicated customers is supported, each provisioned with up to 100 Mbps of bandwidth. The addressing scheme uses a /27 documentation prefix (203.0.113.0/27), subnetted into ten /31 point-to-point allocations. This minimizes address waste and ensures customer isolation.

Private addressing for shared services follows RFC 1918 [17].

5.6 VLAN and Customer Design (Dedicated Services)

5.6.1 Per-Customer VLAN and /31 Allocation

Each dedicated customer is assigned:

- A unique VLAN selected from the range (VLAN 110–120)
- A /31 point-to-point subnet
- A dedicated access port

Customer	VLAN	Gateway (SVI)	Customer IP
1	110	203.0.113.0	203.0.113.1
2	111	203.0.113.2	203.0.113.3
3	112	203.0.113.4	203.0.113.5
⋮	⋮	⋮	⋮
10	119	203.0.113.18	203.0.113.19

Table 4: Customer VLAN and /31 Allocation (Production LLD)

PoC Implementation: The PoC implements a reduced subset of VLANs and /31 links to validate routing behavior without replicating the full customer scale.

5.7 Access Layer Design

Production LLD: The access layer operates at Layer 2. Each customer port is assigned to its dedicated VLAN. Bandwidth policing (100 Mbps) is applied per customer.

Redundant uplinks use LACP-based EtherChannel toward the distribution layer.

PoC Implementation: The PoC uses redundant uplinks with LACP. Bandwidth policing is not implemented; instead, the focus is on validating routing and failover behavior. Bandwidth enforcement is left as future work.

PortFast was not configured in the PoC since the switching topology was minimal and STP convergence was not a focus of the study

5.8 Distribution Layer Design

Production LLD: The distribution layer provides Layer 3 termination for all customer VLANs using SVIs. Redundant LACP trunks connect to the access layer.

PoC Implementation: In the PoC, the distribution layer provides Layer 3 termination for all customer VLANs using SVIs as well. Redundant LACP trunks connect to the access layer.

5.8.1 Uplink from Distribution to Core (LACP + Trunk)

Production LLD: Redundant uplinks between the distribution and core layers are implemented using LACP-based EtherChannel with VLAN trunking. This design provides high availability, increased bandwidth, and efficient transport of aggregated traffic toward the core layer.

PoC Implementation: In the PoC, redundant uplinks and LACP are not implemented; only the routing and failover logic are validated.

5.8.2 iBGP with Core

Production LLD: Internal BGP (iBGP) is established between all the distribution and core layer switches to ensure consistent route propagation and redundancy across the network.

PoC Implementation: The PoC implements a simplified iBGP session between the distribution and core routers to validate internal route propagation.

5.9 Core Layer Design

Production LLD: The core layer aggregates all customer routes and forwards them to the clustered firewall. This hierarchical and modular approach aligns with established best practices for scalable enterprise network architectures [13, 14].

PoC Implementation: The PoC uses a single core router that aggregates routes and forwards traffic to a single firewall/router device.

5.9.1 Downlink from Core to Distribution (LACP + Trunk)

Production LLD: Redundant downlinks from the core to the distribution layer are implemented using LACP-based EtherChannel with VLAN trunking.

PoC Implementation: A single routed link is used without LACP.

5.9.2 Core to Firewall

Production LLD: The core connects to the Active–Standby Check Point cluster using a /31 Layer 3 point-to-point link.

PoC Implementation: A /31 point-to-point link is implemented between the core router and the PoC firewall/router to preserve routing behavior.

5.9.3 iBGP with Distribution

Production LLD: The distribution layer advertises dedicated and shared service routes to the core using iBGP. Static routing is used on the core to forward both service prefixes to the firewall.

PoC Implementation: The PoC implements the same iBGP relationship but with fewer prefixes and simplified static routing.

5.10 Firewall Integration for Dedicated Services

Production LLD: The clustered firewall provides Layer 3 termination for dedicated customer services. Dedicated services use public addressing and do not require NAT. Traffic is steered using Check Point PBR.

PoC Implementation: The PoC uses Cisco IOS PBR to emulate the Check Point routing logic. Dedicated services are routed without NAT, matching the production intent.

5.11 Shared Services

Production LLD: Shared services use private addressing and rely on NAT at the firewall. ISP Blue is the primary path; ISP Red is the backup.

PoC Implementation: The PoC implements private addressing and NAT using Cisco IOS. ISP Blue and ISP Red are simulated using two upstream routers.

5.11.1 Addressing Scheme

Private addressing follows RFC 1918 [17]. The PoC uses the same addressing structure but with fewer VLANs.

5.11.2 Access Layer Design

Production LLD: Shared services are segmented into VLANs. Redundant LACP uplinks connect to the distribution layer.

PoC Implementation: VLAN segmentation is implemented, with redundant uplinks.

5.11.3 Distribution Layer Design

Production LLD: The distribution layer provides Layer 3 gateway functionality and HSRP redundancy.

PoC Implementation: HSRP is not implemented. A single distribution switch provides gateway functionality.

5.11.4 SVIs and HSRP Configuration

Production LLD: HSRP provides first-hop redundancy for shared VLANs.

PoC Implementation: HSRP is omitted. SVIs are configured on a single device.

5.11.5 iBGP with Core

Production LLD: Shared-service VLANs are advertised to the core using iBGP.

PoC Implementation: The PoC advertises the implemented VLANs via iBGP.

5.12 Firewall NAT for Shared Services

Production LLD: Shared services use Hide NAT behind the ISP Blue interface, with a backup NAT rule for ISP Red.

PoC Implementation: Cisco IOS NAT is used to replicate Hide NAT behavior. Backup NAT is implemented using route tracking.

5.13 Firewall Routing, Policy-Based Routing, and BFD Failover

Production LLD: The firewall enforces service-specific routing using static routes, PBR, and BFD for sub-second failover.

PoC Implementation: BFD is not available. IP SLA with object tracking is used to emulate reachability detection and failover.

5.14 Static Route Configuration

Production LLD: Static routes with priorities define primary and backup ISP paths.

PoC Implementation: Equivalent static routes are configured on Cisco IOS using administrative distance.

5.15 Policy-Based Routing

Production LLD: PBR rules enforce service separation based on source networks.

PoC Implementation: Cisco IOS route-maps replicate the same logic.

5.16 BFD and IP Reachability Detection

Production LLD: BFD provides sub-second failure detection.

PoC Implementation: IP SLA probes monitor ISP reachability. Failover occurs within approximately four seconds.

5.17 Failover and Failback Operation

Production LLD: Failover is triggered by BFD. Traffic automatically switches to the backup ISP.

PoC Implementation: Failover is triggered by IP SLA tracking. Traffic returns to the preferred ISP when reachability is restored.

The `verify-availability` keyword ensures that the tracked object only changes state when both the interface and the IP SLA probe fail. This prevents false failovers caused by transient packet loss or momentary probe delays. As a result, failover becomes more stable and avoids unnecessary route flapping.

5.18 Design Adaptation and Platform Selection

The production LLD includes clustered firewalls, VSS/StackWise switching, and BFD. The PoC uses Cisco IOS hardware without these features but preserves all routing logic, service separation, and failover behavior.

Although the syntax differs between Check Point Gaia and Cisco IOS, both platforms support the core mechanisms required by the design, including source-based routing, next-hop enforcement, route tracking, and automatic failover. The objective of the PoC is not to validate vendor-specific firewall functionality, but rather to validate the underlying routing logic, service separation, and failover behavior. Therefore, Cisco IOS provides a suitable platform for evaluating the functional requirements of the proposed architecture under controlled conditions.

5.19 Implementation

The PoC implementation includes:

- VLAN segmentation for dedicated and shared services
- Inter-VLAN routing at the distribution layer
- Internal BGP (iBGP) between distribution and core (AS 6500)
- Policy-Based Routing for service differentiation
- NAT for shared services

- Automatic failover using IP SLA and object tracking

The full equipment list and physical topology are provided in Appendix A, the complete addressing plan is included in Appendix B and all device configurations are available in Appendix C. All configuration steps were aligned with vendor best-practice guidelines, including the Cisco IOS IP SLA documentation and the Cisco Policy-Based Routing configuration guide administration manuals. To clearly demonstrate how each requirement was implemented and validated, Table 5 summarizes the full requirement–design-implementation–result chain.

Requirement	**Design Choice**	**Implementation**	**Test Result**
Service separation	PBR per-service ACLs	Route-map with ACL-FOR-RED/BLUE	100% correct traffic steering
Automatic failover	IP SLA + tracking	SLA 1/2 + track 1/2	4s failover, 0.4% loss
Dual-ISP redundancy	Two upstream links	R1/R2 + static defaults	Symmetric failover behavior
Shared service NAT	NAT-POOL-BLUE	Inside/outside NAT config	Successful translation
Internal routing	iBGP between core/dist	BGP 6500 sessions	Stable route propagation

Table 5: Requirement–Design–Implementation–Result Traceability

6 Results

This chapter presents the results obtained from the proof-of-concept (PoC) implementation of the proposed multihomed network architecture. All tests were performed on the Cisco IOS–based homelab described in Chapter 5.19. The results focus on connectivity verification, Policy-Based Routing (PBR) behavior, ISP failover operation, and baseline performance measurements.

All results were collected from a physical Cisco IOS environment. ISP failures were reproduced through controlled link disconnections and IP SLA target withdrawal to simulate real-world outage conditions in a safe and repeatable manner.

6.1 Connectivity Verification (End-to-End Connectivity)

End-to-end connectivity was verified across all layers of the topology following the initial configuration. ICMP tests were conducted between adjacent and non-adjacent devices to validate static routing, NAT operation, and inter-VLAN routing functionality.

All connectivity tests achieved a 100% success rate under normal operating conditions, as summarized in Table 6.

These results demonstrate that baseline routing, NAT, and inter-VLAN forwarding operated as defined in the Low-Level Design, with no anomalies observed.

Source	Destination	Protocol	Result
DISTRIBUTION	CORE (10.0.0.2)	ICMP	Pass (100%)
DISTRIBUTION	ISP-RED interface (198.51.100.2)	ICMP	Pass (100%)
DISTRIBUTION	ISP-BLUE interface (192.0.2.2)	ICMP	Pass (100%)
DISTRIBUTION	ISP-RED loopback (8.8.8.1)	ICMP	Pass (100%)
DISTRIBUTION	ISP-BLUE loopback (9.9.9.1)	ICMP	Pass (100%)
CORE	ISP-RED (198.51.100.1)	ICMP	Pass (100%)
CORE	ISP-BLUE (192.0.2.1)	ICMP	Pass (100%)
VLAN 110 (203.0.113.0)	ISP-RED loopback (8.8.8.1)	ICMP	Pass (100%)
VLAN 200 (10.10.10.1)	ISP-BLUE loopback (9.9.9.1)	ICMP	Pass (100%)

Table 6: End-to-End Connectivity Results

6.2 Normal Operation — PBR Traffic Steering Results

Policy-Based Routing (PBR) behavior was verified under normal operating conditions with both ISPs active. ICMP tests consisting of 1000 packets were transmitted from dedicated and shared service VLANs toward their respective ISP destinations.

Source VLAN	Source IP	Destination	Expected ISP	Sent	Result
VLAN 110 (Dedicated)	203.0.113.0	8.8.8.1 (ISP-RED)	ISP-RED	1000	100%
VLAN 200 (Shared)	10.10.10.1	9.9.9.1 (ISP-BLUE)	ISP-BLUE	1000	100%

Table 7: Normal Operation PBR Traffic Steering Results

The test results show that traffic originating from the dedicated-service VLAN used ISP-RED, while traffic originating from the shared-service VLAN used ISP-BLUE during normal operation.

6.3 ISP Failover Testing (Test Methodology)

ISP failover was simulated by physically disconnecting the link between the ISP router and the edge router while continuous ICMP traffic was transmitted from the distribution switch. This approach ensured deterministic and repeatable failure conditions, consistent with the methodology described in Chapter 3.

Two failover scenarios were tested:

- **Scenario 1 — ISP-BLUE failure:** Shared service traffic (VLAN 200) was sourced toward the ISP-RED loopback (8.8.8.1) while the ISP-BLUE link was disconnected.
- **Scenario 2 — ISP-RED failure:** Dedicated service traffic (VLAN 110) was sourced toward the ISP-BLUE loopback (9.9.9.1) while the ISP-RED link was disconnected.

Each scenario was executed five times to ensure consistency. Failover time was measured using continuous ICMP probes at one-second intervals, resulting in a measurement uncertainty of approximately ±1 second.

6.4 Failover Results

The results of both failover scenarios are presented in Table 8. Packet loss occurred only during the transition period when the primary ISP became unavailable and the backup route was activated.

Scenario	**Source**	**Destination**	**Sent**	**Success**	**Lost**	**Loss %**	**Failover Time**
ISP-BLUE failure	VLAN 200	8.8.8.1	1000	996	4	0.4%	≈4s
ISP-RED failure	VLAN 110	9.9.9.1	1000	996	4	0.4%	≈4s

Table 8: ISP Failover Test Results

6.5 Observed Failover Sequence

The following sequence of events was observed during both failover scenarios:

1. ISP physical link disconnected.
2. The EDGE-FW interface transitioned to a down state.
3. IP SLA probe returned a timeout.
4. The associated track object transitioned from *Up* to *Down*.
5. PBR `verify-availability` detected the primary next hop as unavailable.
6. PBR redirected traffic to the backup ISP next hop.
7. Traffic resumed via the backup ISP.

This sequence confirms that IP SLA and object tracking operated as intended, providing deterministic failover behavior for both dedicated and shared services.

6.6 Baseline Performance Metrics (UDP Jitter SLA Results)

Baseline performance metrics were collected using Cisco IOS IP SLA UDP jitter operations. Each probe transmitted 100 UDP packets toward the respective ISP gateway.

The successful and failed probe counters shown in Table 9 are cumulative IP SLA statistics collected throughout the entire testing period. The failed probes recorded for ISP-BLUE (SLA 10) and ISP-RED (SLA 11) correspond directly to the periods during which each ISP link was intentionally disconnected during failover testing. These failures are therefore expected and do not indicate instability during normal operation.

Metric	ISP-BLUE (SLA 10)	ISP-RED (SLA 11)
RTT Min/Avg/Max	1/1/1 ms	1/1/1 ms
Jitter S→D Min/Avg/Max	0/0/0 ms	0/0/0 ms
Jitter D→S Min/Avg/Max	0/0/0 ms	0/0/0 ms
One-way Latency S→D	0 ms (NTP required)	0 ms (NTP required)
One-way Latency D→S	0 ms (NTP required)	0 ms (NTP required)
Packet Loss S→D	0	0
Packet Loss D→S	0	0
Out of Sequence Packets	0	0
Tail Drop	0	0
Successful Probes	4	1
Failed Probes	13	3

Table 9: UDP Jitter SLA Baseline Performance Metrics

All completed probes reported zero packet loss and zero jitter, confirming stable baseline performance whenever the ISP links were operational. The probe counters should therefore be interpreted as evidence that IP SLA correctly detected simulated outages rather than as indicators of degraded network performance.

One-way latency values appear as 0 ms because NTP synchronization was not implemented in the homelab. This is expected behavior for IP SLA jitter operations without synchronized clocks.

6.7 Achievement of Design Objectives

Table 10 summarizes the implementation results in relation to the original design objectives.

The results show that all core functional objectives were achieved in the proof-of-concept implementation. The objectives that were not achieved relate exclusively to hardware-dependent redundancy features (VSS, HSRP, clustered firewalls, BFD), which were outside the scope of the homelab environment but remain fully supported by the Low-Level Design.

Overall, the proof-of-concept implementation successfully validated the routing logic, service separation, NAT behavior, and failover mechanisms defined in the Low-Level Design. The measured performance aligns with expectations for IP SLA-based failover and confirms that the proposed architecture can deliver reliable multihomed connectivity under controlled failure conditions.

7 Discussion

This chapter discusses and analyzes the results obtained from the proof-of-concept implementation. The findings are evaluated in relation to the original design objectives, the implementation methodology, and the limitations of the homelab environment. The discussion also reflects on how well the proof-of-concept validated the functional behavior defined in the Low-Level Design and how these results relate to the intended production

Design Objective	Status	Notes
Dual ISP multihomed connectivity	Achieved	Both ISPs connected and operational
Dedicated service routing via ISP-RED	Achieved	PBR verified, 100% success rate
Shared service routing via ISP-BLUE	Achieved	PBR verified, 100% success rate
Automatic ISP failover	Achieved	≈4 seconds, 0.4% packet loss
PBR fallback to backup ISP	Achieved	Verified in both scenarios
NAT for shared services	Achieved	PAT via NAT pool (209.165.1.0/28)
Per-customer VLAN isolation	Achieved	Demonstrated using dedicated and shared VLANs
Performance metric collection	Achieved	UDP jitter SLA on both ISPs
Firewall cluster (Active–Standby)	Not achieved	Hardware and licensing constraints
VSS/StackWise redundancy	Not achieved	Proprietary hardware not available
HSRP first-hop redundancy	Not achieved	Single distribution switch implementation
Sub-second failover via BFD	Not achieved	IP SLA used instead of BFD

Table 10: Design Objectives Achievement Summary

architecture.

7.1 Analysis and Discussion of Results

7.2 Connectivity Analysis

The successful end-to-end connectivity across all layers confirms that the static routing design at the network edge, inter-VLAN routing via SVIs, internal BGP route propagation between the distribution and core layers, and NAT configuration were correctly implemented. The ability to reach both ISP loopback addresses from the distribution layer demonstrates that the edge router correctly forwarded traffic toward both upstream providers and that return routing functioned correctly in both directions.

These results show that the foundational routing and forwarding components behaved deterministically, forming a stable baseline for subsequent evaluation of **Policy-Based Routing** and **failover behavior**.

7.3 PBR Traffic Steering Analysis

Both dedicated and shared service traffic were correctly steered toward their respective ISPs under normal operating conditions, achieving 100% packet delivery with zero packet loss. This validates that the PBR configuration correctly classified and forwarded traffic based on source networks, fulfilling the primary traffic-separation objective of the architecture.

The consistently low RTT values reflect the controlled nature of the homelab environment. Although these values are not representative of real-world ISP performance, they confirm stable forwarding behavior and correct PBR operation.

Overall, the results demonstrate that the architecture achieved strict service separation between dedicated and shared services, one of the core requirements for multihomed enterprise environments.

7.4 Failover Analysis

Both failover scenarios produced identical results, with approximately four lost packets and an estimated failover time of four seconds. The consistency of these results confirms that the failover mechanism operated reliably for both dedicated and shared services.

The failover duration is primarily determined by the configured IP SLA probe frequency and the route-verification process used by the `verify-availability` command. Although the production-oriented design specified BFD for sub-second convergence, the Cisco IOS IP SLA implementation successfully demonstrated automatic failover behavior within the constraints of the proof-of-concept environment.

The temporary ICMP unreachable responses observed during the transition phase confirm that the failover mechanism actively detected the unavailable next hop before rerouting traffic toward the backup ISP.

The bar chart in Figure 7 shows that both ISP failover scenarios produced identical characteristics, with approximately four seconds of convergence time and 0.4% packet loss. This symmetry indicates that the IP SLA tracking and PBR fallback mechanisms behaved consistently, regardless of which ISP became unavailable.

The observed failover time is consistent with the configured IP SLA probe interval of five seconds. Because failure detection depends on periodic probe execution and object tracking, convergence occurs on the order of several seconds rather than the sub-second performance typically achieved with BFD.

Although the PoC was not designed as a statistical study, the repeated failover tests showed minimal variance. Failover time ranged between 3–5 seconds across all runs, and packet loss varied between 3–5 packets per 1000. This narrow spread indicates that the results are consistent and not the outcome of a single anomalous measurement.

7.5 Performance Metrics Analysis

The UDP jitter SLA measurements showed identical and stable performance characteristics for both ISP links under normal operating conditions. RTT values remained consistently low, while jitter and packet loss remained at zero throughout baseline testing.

These results are expected in a controlled lab environment with direct physical connections and no external congestion or propagation delay. In a production environment,

ISP performance characteristics would vary depending on provider infrastructure, routing paths, and network load.

The one-way latency measurements returned zero because Network Time Protocol (NTP) synchronization was not configured between devices. Accurate one-way delay measurements require synchronized clocks between the IP SLA source and responder devices.

Although the absolute values are not representative of real-world ISP performance, the stability of the measurements confirms that the architecture introduces no additional jitter or latency under normal conditions.

7.6 Overall Evaluation

The results demonstrate that the primary objectives of the thesis were successfully achieved within the limitations of the homelab environment. Dual-ISP multihoming, service separation, Policy-Based Routing, NAT, and automatic failover were all validated through practical testing.

The objectives that were not achieved relate exclusively to advanced redundancy features requiring proprietary hardware or commercial firewall platforms that were unavailable within the project scope. These limitations do not invalidate the proof-of-concept implementation but instead highlight the difference between a laboratory validation environment and a production-grade deployment.

Overall, the proof-of-concept successfully validated the routing logic, failover behavior, and service differentiation defined in the Low-Level Design, despite the absence of production-grade redundancy features. The results demonstrate that the routing logic, PBR rules, and failover mechanisms are functionally sound and behave predictably under controlled conditions.

The results indicate that all primary functional requirements identified in Chapter 4 were satisfied. High availability was demonstrated through successful dual-ISP failover, service separation was achieved through PBR, and centralized traffic control was validated through deterministic routing behavior. The only requirements not fully implemented were those dependent on hardware-specific redundancy technologies such as firewall clustering, HSRP, and VSS/StackWise. These limitations affected infrastructure redundancy but did not affect the validation of the routing architecture itself.

7.7 Project Method Discussion

The project followed a proof-of-concept methodology in which the production-oriented Low-Level Design was adapted into a simplified Cisco IOS-based implementation suitable for a controlled homelab environment.

The use of Boson NetSim and Cisco IOS devices provided a reproducible and stable testing platform while avoiding the licensing and hardware requirements associated with Check Point Gaia firewall clusters and enterprise switching platforms. This approach allowed the project to focus on validating the functional routing behavior of the proposed architecture rather than vendor-specific implementation details.

Although several production-oriented redundancy mechanisms were omitted including firewall clustering, VSS/StackWise, and HSRP, the adapted implementation preserved the core architectural behavior defined in the Low-Level Design. The methodology, therefore, achieved the primary project milestones related to multihoming, traffic steering, failover, and service separation.

The controlled homelab environment also ensured reproducibility, allowing repeated failover tests under identical conditions, which are essential requirements for quantitative evaluation.

7.8 Scientific Discussion

The findings of this thesis demonstrate that multihomed enterprise network behavior can be successfully validated in a simplified proof-of-concept environment while still preserving the functional intent of a production-oriented architecture. The results show that the core mechanisms of multihoming (Policy-Based Routing, dual-ISP failover, NAT behavior, and internal route propagation) can be reproduced using standard Cisco IOS functionality without requiring specialized carrier-grade equipment.

The implementation confirms that Policy-Based Routing can effectively enforce service separation between dedicated public services and shared private services in a dual-ISP environment. The failover tests further demonstrate that dynamic ISP selection and automatic rerouting can be achieved using IP SLA and object tracking, even though the original design specified BFD for sub-second convergence.

From a scientific perspective, the project contributes a practical demonstration of how enterprise multihoming concepts can be implemented and validated within constrained educational or laboratory environments. This directly addresses the scientific gap identified in Section 2.4, where previous studies focused on large-scale Internet measurements or simulation-based evaluations rather than practical, reproducible proof-of-concept implementations.

The findings support prior research on multihoming and routing convergence, particularly the work of Akella et al. and Basit et al., which emphasize the importance of routing policy design and failure-detection mechanisms. The results also illustrate the trade-offs between production-grade architectures and simplified PoC implementations, showing how routing logic can be preserved even when redundancy features are omitted. Unlike many previous studies that rely primarily on simulation or large-scale Internet measurements, this thesis demonstrates that multihoming concepts can be validated using commercially available networking equipment in a reproducible homelab environment. This provides practical insight into the implementation challenges associated with Policy-Based Routing, NAT, and failover mechanisms, complementing the theoretical findings reported in earlier research.

From a validity perspective, the internal validity of the results is strong because all tests were performed under controlled and repeatable conditions. However, the external validity is limited by the simplified homelab environment, which does not replicate real ISP latency, congestion, or routing complexity. Therefore, the results should be interpreted as functional validation rather than predictive indicators of production-grade performance.

The originality of this thesis lies primarily in the practical proof-of-concept implementation and empirical measurement of multihoming behavior using constrained hardware. While the underlying technologies are well-established, the contribution of this work is the demonstration of how production-oriented routing logic can be validated in an educational lab environment.

The results can be partially generalized to real enterprise networks. The routing logic, PBR behavior, and failover mechanisms follow standard networking principles and therefore translate directly to production environments. However, the absolute performance

values, such as latency, jitter, and failover time, cannot be generalized because real ISP networks introduce congestion, asymmetric routing, and variable delay. The PoC therefore validates the design principles but not the exact performance characteristics of a production deployment.

7.9 Study Limitations

Several limitations should be considered when interpreting the results. First, the implementation was performed in a controlled homelab environment rather than a production network. As a result, measured latency, jitter, and packet-loss values do not reflect real ISP conditions. Second, the PoC omitted production-grade redundancy mechanisms such as firewall clustering, HSRP, VSS/StackWise, and BFD due to hardware and licensing constraints. Third, performance testing focused primarily on failover behavior and did not include throughput measurements, long-term stability testing, or congestion scenarios. Consequently, the results should be interpreted as validation of architectural functionality rather than a complete performance evaluation of a production deployment.

7.10 Recommendations

Based on the results obtained, the following recommendations are made for a production deployment of this architecture:

- **Replace IP SLA with BFD** to achieve sub-second ISP failure detection and faster failover convergence.
- **Deploy a redundant firewall cluster** to eliminate the edge firewall as a single point of failure.
- **Implement VSS or StackWise** at the switching layers to improve infrastructure redundancy and simplify management.
- **Configure HSRP or VRRP** on distribution-layer SVIs to provide first-hop gateway redundancy for shared service VLANs.
- **Deploy NTP synchronization** across all devices to support accurate one-way latency measurements.
- **Expand performance testing** to include throughput measurements, congestion testing, and long-duration stability monitoring.
- **Evaluate the architecture using real ISP circuits** or emulated WAN delay to assess performance under realistic propagation and congestion conditions.

These recommendations align the PoC implementation with the full production Low-Level Design and address the hardware-dependent features that were outside the scope of the homelab environment.

7.11 Ethical and Societal Discussion

This project focuses on improving network availability, service continuity, and traffic separation in enterprise multihomed environments. Reliable network connectivity is essential for modern organizations because outages may affect communication, business operations, and access to digital services. From a societal perspective, resilient network infrastructures contribute to the stability of digital services that support healthcare, education, commerce, and public administration.

The implementation of automatic failover mechanisms contributes to improved service availability and operational resilience by reducing downtime during ISP failures. This aligns with responsible engineering practices, where reliability and continuity are considered essential components of system design.

The project was implemented entirely within a controlled simulation and homelab environment. This approach minimized financial costs, hardware requirements, and operational risks while allowing repeated testing without affecting production systems or users. No personal user data was processed during the implementation or testing phases. The exclusive use of documentation prefixes and isolated lab infrastructure ensured that no production traffic or sensitive data was exposed.

From an ethical perspective, the project demonstrates responsible engineering by validating network behavior in a safe and reproducible environment before theoretical deployment into production infrastructure. The methodology avoided any risk to real users, real networks, or sensitive information, ensuring that the work adhered to ethical standards for network experimentation.

Beyond these general reflections, network engineering decisions often carry practical ethical consequences that affect availability, privacy, and operational safety. To make these considerations more explicit, Table 11 presents a structured overview of the main ethical and operational risks associated with multihomed network architectures.

Risk Category	**Potential Consequence**
Availability	Misconfigured failover causing prolonged outage during ISP failure.
Privacy	Incorrect NAT or routing exposing internal IPs or services to the Internet.
Security Risk	Traffic bypassing firewall inspection during failover due to incorrect PBR rules.
Operational Risk	Human error during configuration leading to service disruption.
Cost/Energy	Redundant devices increasing operational cost and energy consumption.
Misconfiguration Risk	Sensitive traffic routed over the wrong ISP, violating policy or compliance.

Table 11: Ethical and operational risk matrix for multihomed network design

Real incidents illustrate these risks. For example, several organizations have experienced outages when misconfigured failover rules created routing loops during ISP transitions. In other cases, incorrect NAT rules exposed internal services to the public Internet,

resulting in data leaks and security breaches. These examples highlight the ethical responsibility to validate routing logic thoroughly before deployment.

7.12 Operational Risks and Rollback Considerations

The deployment of a multihomed architecture introduces several operational risks. Misconfigured PBR rules may route traffic to the wrong ISP, incorrect NAT rules may expose internal services, and IP SLA thresholds may trigger unnecessary failovers. To mitigate these risks, a staged deployment plan should be used, beginning with monitoring-only IP SLA, followed by shadow PBR policies, and finally full activation. A rollback plan should include restoring the previous default route configuration, disabling PBR route-maps, and reverting NAT rules to the last known working state.

8 Conclusion

This section summarizes the main findings of the thesis and evaluates the extent to which the project achieved its objectives.

8.1 Goal Achievement

This thesis investigated how a multihomed secondary Internet network can be designed and implemented to provide service separation, high availability, and automatic failover using dual ISP connectivity. The project was based on a real-world client scenario at 2E Engineering, where the objective was to redesign a secondary Internet infrastructure capable of supporting both dedicated and shared customer services while maintaining resilience during ISP outages.

The implementation was delivered as a functional proof-of-concept using Cisco IOS hardware, preserving the routing intent and service behavior of the production Low-Level Design. Although the PoC did not include all production-grade redundancy mechanisms, it successfully validated the core routing logic and failover behavior of the proposed architecture.

The architecture followed a hierarchical three-tier model consisting of access, distribution, and core layers connected to two upstream ISPs. Policy-Based Routing was used to enforce service-based traffic steering, ensuring that dedicated customer traffic was routed through ISP-RED while shared services used ISP-BLUE as the preferred path. Automatic failover was implemented using IP SLA tracking and route-verification mechanisms.

The results demonstrated that multihoming can be implemented effectively even in constrained environments, provided that routing intent and service separation are preserved. Under normal operating conditions, both dedicated and shared services achieved 100% packet delivery with stable, low-latency connectivity. During simulated ISP failures, traffic automatically failed over to the backup ISP with approximately four seconds of convergence time and only 0.4% packet loss. These findings confirm that the combination of PBR and IP SLA provides a reliable mechanism for enforcing service-based routing and automated failover in dual-ISP environments.

The research questions defined in Chapter 1 were successfully answered. The study demonstrated how multihomed networks can achieve high availability through dual ISP connectivity and automated failover mechanisms. It also showed that PBR combined with IP SLA tracking can effectively enforce service-based access policies across multiple

ISPs while maintaining stable connectivity for both dedicated and shared services. All research questions were addressed through empirical testing, demonstrating that the proposed architecture meets the functional requirements of a multihomed secondary Internet network. Furthermore, the evaluation presented in Table 10 shows that all core functional design objectives were achieved. The objectives that were not achieved were limited to hardware-dependent redundancy features, including firewall clustering, HSRP, VSS/StackWise, and BFD-based failover, which were outside the scope of the proof-of-concept implementation.

8.2 Limitations

It is important to distinguish between the customer-approved Low-Level Design, which specifies a production-grade architecture with clustered firewalls and switch redundancy, and the simplified proof-of-concept implementation, which validates only the routing logic and failover behavior under controlled conditions.

While the proof-of-concept validates the routing logic and failover behavior, it does not fully represent the performance, scale, or operational complexity of a production deployment.

Although several adaptations were required due to hardware and software limitations, including replacing Check Point Gaia clustering with Cisco IOS-based routing, the core functionality of the design was preserved. These adaptations do not affect the functional validation of the routing logic and failover mechanisms; however, they limit the extent to which the measured performance can be generalized to production environments.

8.3 Scientific and Practical Contribution

The project also contributes practical insights into Cisco IOS multihoming behavior. In particular, the implementation demonstrated that IP SLA can serve as a functional alternative to BFD in environments where BFD is unavailable, albeit with slower failover performance. Additionally, the project identified the operational requirement for static host routes when using PBR `verify-availability`, an important implementation detail for reliable failover operation in Cisco IOS environments. From a scientific perspective, the thesis contributes an empirical proof-of-concept demonstrating how enterprise multihoming principles can be validated using constrained educational hardware while preserving the functional behavior of a production-oriented architecture.

Overall, the project confirms that the proposed architecture is functionally suitable for multihomed secondary Internet deployments requiring service differentiation, traffic steering, and automated ISP redundancy.

From a broader perspective, the project contributes a practical and reproducible demonstration of multihoming behavior using widely available networking equipment. By validating routing logic, failover mechanisms, and service separation in a constrained environment, the thesis provides a methodological blueprint for evaluating enterprise network designs without requiring full production infrastructure.

In summary, the project demonstrates that a carefully designed multihomed architecture can deliver reliable service continuity, predictable failover behavior, and clear service separation, even when implemented in a simplified proof-of-concept environment. These findings provide a solid foundation for future production deployment and further research into enterprise multihoming strategies.

8.4 Future Work

Several areas for future work were identified during the project:

- **Implement the full production design** using Check Point Gaia R81.20 with Active–Standby firewall clustering and BFD-based failover to evaluate sub-second convergence performance.
- **Evaluate the architecture using real ISP circuits** or WAN emulation tools capable of introducing realistic latency, jitter, and packet-loss conditions.
- **Implement Layer 2 access security mechanisms** such as IP Source Guard to reduce the risk of accidental customer misconfiguration.
- **Deploy switch-level redundancy mechanisms** such as VSS, StackWise, and HSRP using enterprise-grade Cisco hardware to validate full infrastructure redundancy.
- **Scale the architecture to the full ten-customer deployment** defined in the Low-Level Design, including load testing.
- **Investigate SD-WAN integration** to compare traditional PBR-based multihoming with modern SD-WAN approaches in terms of failover performance, application-aware routing, and operational complexity.
- **Implement QoS and traffic policing mechanisms** to provide per-customer bandwidth guarantees and service prioritization.
- **Investigate additional routing and security hardening measures**, including route summarization, Null0 discard routes, ACLs, and uRPF, to further improve scalability and security in production deployments.

8.5 Future Work Example

One particularly valuable extension of this project would be a direct comparison between IP SLA and Bidirectional Forwarding Detection (BFD) for ISP failover detection and convergence performance. This could be approached by implementing BFD sessions between the edge router and both ISP routers using Cisco IOS 15.5, which supports BFD functionality on the available hardware.

Controlled ISP failure simulations could then be repeated while measuring failover time, packet loss, jitter, and routing convergence behavior under identical test conditions. The collected measurements could be compared directly with the IP SLA baseline established in this thesis to quantify the performance improvements achievable with BFD. Such a study would provide useful empirical data for organizations evaluating failover technologies in multihomed enterprise networks.

Appendix

The original Low-Level Design (LLD) was based on a production-oriented architecture consisting of redundant switching layers, a clustered Check Point firewall platform, and

advanced routing and failover mechanisms. Due to time, hardware, and licensing constraints, the production firewall solution was replaced with a Cisco IOS–based implementation in a homelab environment.

Although the implementation differs from the customer-approved production design in terms of hardware and redundancy features, the routing logic, traffic-engineering policies, NAT behavior, and failover mechanisms remain functionally equivalent for the purposes of validation. These adaptations resulted in a functional proof-of-concept implementation that preserved the routing intent, service separation, and failover logic of the original LLD. The simplified implementation environment, therefore, validates the operational behavior of the intended production design while remaining suitable for academic testing and reproducibility.

A Physical Equipment Specifications

Device	Model	Role	IOS Version	Interfaces Used
R1	Cisco 2911	ISP-RED	15.1	Gig0/1, Loopback0
R2	Cisco 2911	ISP-BLUE	15.1	Gig0/0, Loopback1
R3	Cisco 2911	CORE	15.1	Gig0/0, Gig0/1
R4	Cisco 2911	EDGE-FW	15.5	Gig0/0, Gig0/1, Gig0/2
SW1	Cisco 3560	DISTRIBUTION	12.2	Fast0/1, Fast0/15–16
SW2	Cisco 3560	ACCESS	12.2	Fast0/15–16, Fast0/23–24

Table 12: Physical Equipment Specifications

The EDGE-FW device runs IOS 15.5 to support IP SLA tracking and PBR `verify-availability`, which are required for the failover mechanism. EDGE-FW is a Cisco IOS router performing firewall-edge functions in the PoC and does not represent a full stateful firewall implementation.

A.1 Physical Connection Table

Table 13 summarizes the physical cabling used in the proof-of-concept implementation, including routed links, trunk connections, and end-host access ports.

The physical topology reflects a simplified version of the production architecture, preserving logical roles while reducing hardware complexity. The trunk between the access and distribution switches was implemented as an LACP EtherChannel to provide additional bandwidth and demonstrate switch interconnection redundancy concepts, although switch-level failover was outside the scope of the PoC.

B IP Addressing Plan

This section summarizes the IP addressing scheme used in the proof-of-concept implementation, including infrastructure links, customer VLANs, NAT address pools, and ISP test networks.

Device A	Port A	Device B	Port B	Link Type
ISP-RED (R1)	Gig0/1	EDGE-FW (R4)	Gig0/1	Routed L3
ISP-BLUE (R2)	Gig0/0	EDGE-FW (R4)	Gig0/0	Routed L3
EDGE-FW (R4)	Gig0/2	CORE (R3)	Gig0/1	Routed L3
CORE (R3)	Gig0/0	DISTRIBUTION (SW1)	Fast0/1	Routed L3
DISTRIBUTION (SW1)	Fast0/15	ACCESS (SW2)	Fast0/15	802.1Q Trunk
DISTRIBUTION (SW1)	Fast0/16	ACCESS (SW2)	Fast0/16	802.1Q Trunk
ACCESS (SW2)	Fast0/23	Shared-PC	NIC	Access VLAN 200
ACCESS (SW2)	Fast0/24	Dedicated-PC	NIC	Access VLAN 110

Table 13: Physical Connection Table

B.1 Point-to-Point Infrastructure Links

Link	Subnet	Device A	IP A	IP B
ISP-RED ↔ EDGE	198.51.100.0/30	R1 Gig0/1	198.51.100.1	198.51.100.2
ISP-BLUE ↔ EDGE	192.0.2.0/30	R2 Gig0/0	192.0.2.1	192.0.2.2
EDGE ↔ CORE	10.0.0.0/31	R4 Gig0/2	10.0.0.0	10.0.0.1
CORE ↔ DIST	10.0.0.2/31	R3 Gig0/0	10.0.0.2	10.0.0.3

Table 14: Point-to-Point Link Addressing

Table 14 shows the addressing used on routed infrastructure links between the ISP routers, EDGE-FW, CORE, and DISTRIBUTION devices.

All public IP ranges follow the documentation prefix guidelines defined in RFC 5737, ensuring safe use within a lab environment.

B.2 ISP Loopback Addresses

Device	Interface	IP Address
ISP-RED (R1)	Loopback0	8.8.8.1/24
ISP-BLUE (R2)	Loopback1	9.9.9.1/32

Table 15: ISP Loopback Addresses

The ISP loopback addresses shown in Table 15 were used as stable reachability targets during connectivity and failover testing.

Note: The ISP loopback and NAT pool addresses use non-RFC public IPs for testing purposes in the homelab environment. As stated in Chapter 5, these do not represent real customer or provider addresses.

Pool Name	Subnet	Purpose
NAT-POOL-BLUE	209.165.1.0/28	Shared service NAT translation

Table 16: NAT Pool Addressing

B.3 NAT Pool

The shared-service networks use the NAT pool shown in Table 16 when accessing external networks through ISP-BLUE.

B.4 VLAN and SVI Addressing

VLAN	Name	Service Type	Subnet	SVI Gateway
110	Dedicated-1	Dedicated	203.0.113.0/31	203.0.113.0
111	Dedicated-2	Dedicated	203.0.113.2/31	203.0.113.2
200	Shared-1	Shared	10.10.10.0/24	10.10.10.1
210	Shared-2	Shared	10.20.20.0/24	10.20.20.1

Table 17: VLAN and SVI Addressing

Table 17 summarizes the VLAN assignments, service classifications, and corresponding SVI gateway addresses used in the proof-of-concept implementation.

B.5 Dedicated Customer Addressing — Full Scale Design

Dedicated customer VLANs use /31 addressing in accordance with RFC 3021 [18] to conserve public address space.

Customer	VLAN	SVI Gateway	Customer IP
1	110	203.0.113.0	203.0.113.1
2	111	203.0.113.2	203.0.113.3
3	112	203.0.113.4	203.0.113.5
4	113	203.0.113.6	203.0.113.7
5	114	203.0.113.8	203.0.113.9
6	115	203.0.113.10	203.0.113.11
7	116	203.0.113.12	203.0.113.13
8	117	203.0.113.14	203.0.113.15
9	118	203.0.113.16	203.0.113.17
10	119	203.0.113.18	203.0.113.19

Table 18: Full-Scale Dedicated Customer VLAN and /31 Allocation

Although only two customer VLANs were implemented in the proof-of-concept, Table 18 illustrates how the design scales to the full ten-customer deployment specified in

the Low-Level Design.

B.6 Reproducibility Notes

The homelab implementation is fully reproducible using the equipment and addressing plans listed in this appendix. All configurations were tested using standard Cisco IOS commands without requiring proprietary features. The design can also be reproduced in virtual environments such as GNS3 or CML, provided that IP SLA, PBR, and basic routing features are supported.

C Device Configurations

This appendix contains the complete Cisco IOS configurations used in the proof-of-concept implementation. Only configuration elements relevant to routing, Policy-Based Routing, NAT, IP SLA, and failover behavior are included. Default system output and unrelated interface configurations have been omitted for clarity.

C.1 ISP-RED (R1) Configuration

The ISP-RED router simulates the primary upstream provider. Its configuration includes a loopback test network, a point-to-point link to the edge firewall, static routes toward customer prefixes, and the IP SLA responder used for failover verification.

```
hostname ISP-RED
!
interface Loopback0
 description ISP-RED-Test-Network
 ip address 8.8.8.1 255.255.255.0
!
interface GigabitEthernet0/1
 description Link-to-EDGE
 ip address 198.51.100.1 255.255.255.252
 no shutdown
!
ip route 203.0.113.0 255.255.255.224 198.51.100.2
ip route 209.165.1.0 255.255.255.240 198.51.100.2
!
ip sla responder
```

C.2 ISP-BLUE (R2) Configuration

The ISP-BLUE router simulates the secondary upstream provider. Its configuration mirrors ISP-RED, including a loopback test network, static routes toward customer prefixes, and an IP SLA responder for failover testing.

```
hostname ISP-BLUE
!
interface Loopback1
 description ISP-BLUE-Test-Network
```

```
 ip address 9.9.9.1 255.255.255.255
!
interface GigabitEthernet0/0
 description Link-to-EDGE
 ip address 192.0.2.1 255.255.255.252
 no shutdown
!
ip route 203.0.113.0 255.255.255.224 192.0.2.2
ip route 209.165.1.0 255.255.255.240 192.0.2.2
!
ip sla responder
```

C.3 EDGE-FW (R4) Configuration

The EDGE-FW router implements Policy-Based Routing, NAT, IP SLA tracking, and the failover logic that selects between ISP-RED and ISP-BLUE. This is the central device responsible for service separation and automated failover.

```
hostname EDGE-FW
!
track 1 ip sla 1 reachability
track 2 ip sla 2 reachability
!
interface GigabitEthernet0/0
 description Link-to-ISP-BLUE
 ip address 192.0.2.2 255.255.255.252
 ip nat outside
 no shutdown
!
interface GigabitEthernet0/1
 description Link-to-ISP-RED
 ip address 198.51.100.2 255.255.255.252
 ip nat outside
 no shutdown
!
interface GigabitEthernet0/2
 description Link-to-CORE
 ip address 10.0.0.0 255.255.255.254
 ip nat inside
 ip policy route-map PBR_ISP_SELECTION
 no shutdown
!
ip nat pool NAT-POOL-BLUE 209.165.1.1 209.165.1.14 netmask 255.255.255.240
!
ip access-list extended NAT-TO-BLUE
 permit ip 10.10.10.0 0.0.0.255 any
 permit ip 10.20.20.0 0.0.0.255 any
!
ip nat inside source list NAT-TO-BLUE pool NAT-POOL-BLUE overload
```

```
!
ip route 10.10.10.0 255.255.255.0 10.0.0.1
ip route 10.20.20.0 255.255.255.0 10.0.0.1
ip route 203.0.113.0 255.255.255.224 10.0.0.1
!
ip route 0.0.0.0 0.0.0.0 192.0.2.1 10
ip route 0.0.0.0 0.0.0.0 198.51.100.1 20
!
ip access-list extended ACL-FOR-RED
 permit ip 203.0.113.0 0.0.0.31 any
!
ip access-list extended ACL-FOR-BLUE
 permit ip 10.10.10.0 0.0.0.255 any
 permit ip 10.20.20.0 0.0.0.255 any
!
route-map PBR_ISP_SELECTION permit 10
 match ip address ACL-FOR-RED
 set ip next-hop verify-availability 198.51.100.1 1 track 1
 set ip next-hop verify-availability 192.0.2.1 2 track 2
!
route-map PBR_ISP_SELECTION permit 20
 match ip address ACL-FOR-BLUE
 set ip next-hop verify-availability 192.0.2.1 1 track 2
 set ip next-hop verify-availability 198.51.100.1 2 track 1
!
ip sla 1
 icmp-echo 198.51.100.1
 frequency 5
ip sla schedule 1 life forever start-time now
!
ip sla 2
 icmp-echo 192.0.2.1
 frequency 5
ip sla schedule 2 life forever start-time now
```

C.4 CORE (R3) Configuration

The CORE router provides internal routing between the edge firewall and the distribution layer. It also participates in internal BGP to advertise customer and shared service networks. Internal BGP was used to propagate customer and service networks between the distribution and core layers, reflecting the routing model defined in the production Low-Level Design. External ISP connectivity was handled through static routing and Policy-Based Routing at the edge.

```
hostname CORE
!
interface GigabitEthernet0/1
 description Link-to-EDGE
 ip address 10.0.0.1 255.255.255.254
```

```
 no shutdown
!
interface GigabitEthernet0/0
 description Link-to-DISTRIBUTION
 ip address 10.0.0.2 255.255.255.254
 no shutdown
!
ip route 0.0.0.0 0.0.0.0 10.0.0.0
!
router bgp 6500
 bgp log-neighbor-changes
 neighbor 10.0.0.3 remote-as 6500
 neighbor 10.0.0.3 description Distribution-Switch
 !
 address-family ipv4
  neighbor 10.0.0.3 activate
  network 203.0.113.0 mask 255.255.255.224
  network 10.10.10.0 mask 255.255.255.0
  network 10.20.20.0 mask 255.255.255.0
 exit-address-family
```

C.5 DISTRIBUTION (SW1) Configuration

The distribution switch performs inter-VLAN routing for dedicated and shared services, provides SVI gateways, and participates in internal BGP to propagate routes toward the core.

```
hostname DISTRIBUTION
!
ip routing
!
interface FastEthernet0/1
 no switchport
 description Link-to-CORE
 ip address 10.0.0.3 255.255.255.254
 no shutdown
!
vlan 110
 name Dedicated-1
!
vlan 111
 name Dedicated-2
!
vlan 200
 name Shared-1
!
vlan 210
 name Shared-2
!
```

```
interface Vlan110
 description Dedicated-1-SVI
 ip address 203.0.113.0 255.255.255.254
 no shutdown
!
interface Vlan111
 description Dedicated-2-SVI
 ip address 203.0.113.2 255.255.255.254
 no shutdown
!
interface Vlan200
 description Shared-1-SVI
 ip address 10.10.10.1 255.255.255.0
 no shutdown
!
interface Vlan210
 description Shared-2-SVI
 ip address 10.20.20.1 255.255.255.0
 no shutdown
!
interface FastEthernet0/15
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 110,111,200,210
 channel-group 1 mode active
 no shutdown
!
interface FastEthernet0/16
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 110,111,200,210
 channel-group 1 mode active
 no shutdown
!
interface Port-channel1
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 110,111,200,210
!
router bgp 6500
 bgp log-neighbor-changes
 neighbor 10.0.0.2 remote-as 6500
 neighbor 10.0.0.2 description Core-Router
 !
 address-family ipv4
  neighbor 10.0.0.2 activate
  network 203.0.113.0 mask 255.255.255.224
  network 10.10.10.0 mask 255.255.255.0
```

```
  network 10.20.20.0 mask 255.255.255.0
 exit-address-family
```

C.6 ACCESS (SW2) Configuration

The access switch provides VLAN access for end hosts and forms a port-channel trunk toward the distribution layer to carry dedicated and shared service VLANs.

```
hostname ACCESS
!
vlan 110
 name Dedicated-1
!
vlan 111
 name Dedicated-2
!
vlan 200
 name Shared-1
!
vlan 210
 name Shared-2
!
interface FastEthernet0/15
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 110,111,200,210
 channel-group 1 mode active
 no shutdown
!
interface FastEthernet0/16
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 110,111,200,210
 channel-group 1 mode active
 no shutdown
!
interface Port-channel1
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 110,111,200,210
!
interface FastEthernet0/23
 switchport mode access
 switchport access vlan 200
 description Shared-PC
 no shutdown
!
interface FastEthernet0/24
 switchport mode access
```

```
switchport access vlan 110
description Dedicated-PC
no shutdown
```

D Project Timeline

Table 19 presents the project timeline followed during the thesis. The schedule outlines the major project phases, including literature review, design, implementation, testing, analysis, and report writing. The timeline was used as a planning tool throughout the project and was adjusted where necessary to accommodate implementation challenges and testing activities.

Week	**Activity**	**Description**
1–2	Project planning	Analyze client requirements, define scope, write Introduction (Chapter 1)
3–4	Pre-study	Document existing network, identify hardware constraints, begin Theoretical Background (Chapters 2 and 4)
5–6	Design phase	Develop Low-Level Design, service separation model, routing architecture, write Methodology (Chapter 3) and start Design (Chapter 5)
7	Mid-term presentation	Present preliminary design and diagrams, incorporate feedback into Chapters 4 and 5
8	Implementation	Build proof-of-concept using real hardware, configure routing, NAT, VLANs, and PBR (Chapter 5)
9	Testing & evaluation	Simulate ISP failures, verify dedicated/shared services, measure failover time, packet loss, and stability (Chapter 6)
10	Draft report	Compile results, generate diagrams, write Discussion and begin Conclusion (Chapters 7 and 8), submit draft for opposition
11	Final report, opposition and presentation	Revise report, finalize diagrams, perform opposition, present findings, submit final thesis

References

[1] NetworkLessons.com. (n.d.) Bgp topology examples (single-homed, dual-homed, and multihomed designs). Accessed: 2026-04-27. [Online]. Available: https://networklessons.com/bgp

[2] A. Lamberti. (2024) How to monitor internet multihoming networks: From a to z. Accessed: 2026-04-27. [Online]. Available: https://obkio.com/blog/how-to-monitor-internet-multihoming/

[3] A. Akella, S. Seshan, and A. Shaikh, "A comparison of overlay routing and multihoming route control," *IEEE/ACM Transactions on Networking*, 2004, demonstrates that multihoming improves availability and performance compared to single-homed networks.

[4] Z. Basit, M. Khan, and I. Ali, "Performance analysis of ospf and eigrp convergence through ipsec tunnel using multi-homing bgp connection," *International Journal of Computer Networks and Communications*, 2022, evaluates routing performance and failover behavior in multihomed environments.

[5] C. Raiciu, C. Paasch, S. Barre *et al.*, "Improving datacenter performance and robustness with multipath tcp," *ACM SIGCOMM Computer Communication Review*, 2011.

[6] W. Shi, J. Li *et al.*, "Software-defined wan: Architecture and applications," *IEEE Communications Magazine*, 2017.

[7] H. Shin, L. Guo, and L. Qiu, "The impact of multihoming on network reliability and stability," *IEEE Communications Letters*, 2005, analyzes how multihoming affects routing stability and reliability.

[8] Anonymous, "Multihoming: A comprehensive review," *Journal of Network and Computer Applications*, 2013, provides an overview of multihoming architectures and resilience benefits.

[9] Y. Rekhter and T. Li, *BGP*. O'Reilly Media, 2002.

[10] R. Bennesby and N. Fonseca, "A survey on approaches to reduce bgp interdomain routing convergence delay," *Computer Communications*, 2017, discusses BGP convergence challenges and their impact on reliability.

[11] T. Anderson, H. Balakrishnan *et al.*, "Multihoming in ip networks: A survey," *Computer Networks*, 2013.

[12] R. Bush and T. Griffin, "Bgp convergence issues and performance implications," *IEEE Network*, 2004.

[13] L. Corral, "Network infrastructure high-level design," 2020. [Online]. Available: https://ittn-018.lsst.io/

[14] E. I. Technology, "Low-level network design," 2023. [Online]. Available: https://eal-it-technology.github.io/Network-design/lld.html

[15] J. Arkko and S. Bradner, "Ipv4 address blocks reserved for documentation," RFC 5737, 2010. [Online]. Available: https://www.rfc-editor.org/rfc/rfc5737

[16] *Check Point Gaia R81.20 Policy-Based Routing Guide*, Check Point Software Technologies, 2023. [Online]. Available: https://support.checkpoint.com

[17] Y. Rekhter, B. Moskowitz, D. Karrenberg, G. J. de Groot, and E. Lear, "Rfc 1918: Address allocation for private internets," Internet Engineering Task Force, 1996. [Online]. Available: https://www.rfc-editor.org/rfc/rfc1918

[18] P. Retana and A. White, "Using 31-bit prefixes on ipv4 point-to-point links," Internet Engineering Task Force (IETF), RFC 3021, Dec. 2000, best Current Practice. [Online]. Available: https://www.rfc-editor.org/rfc/rfc3021

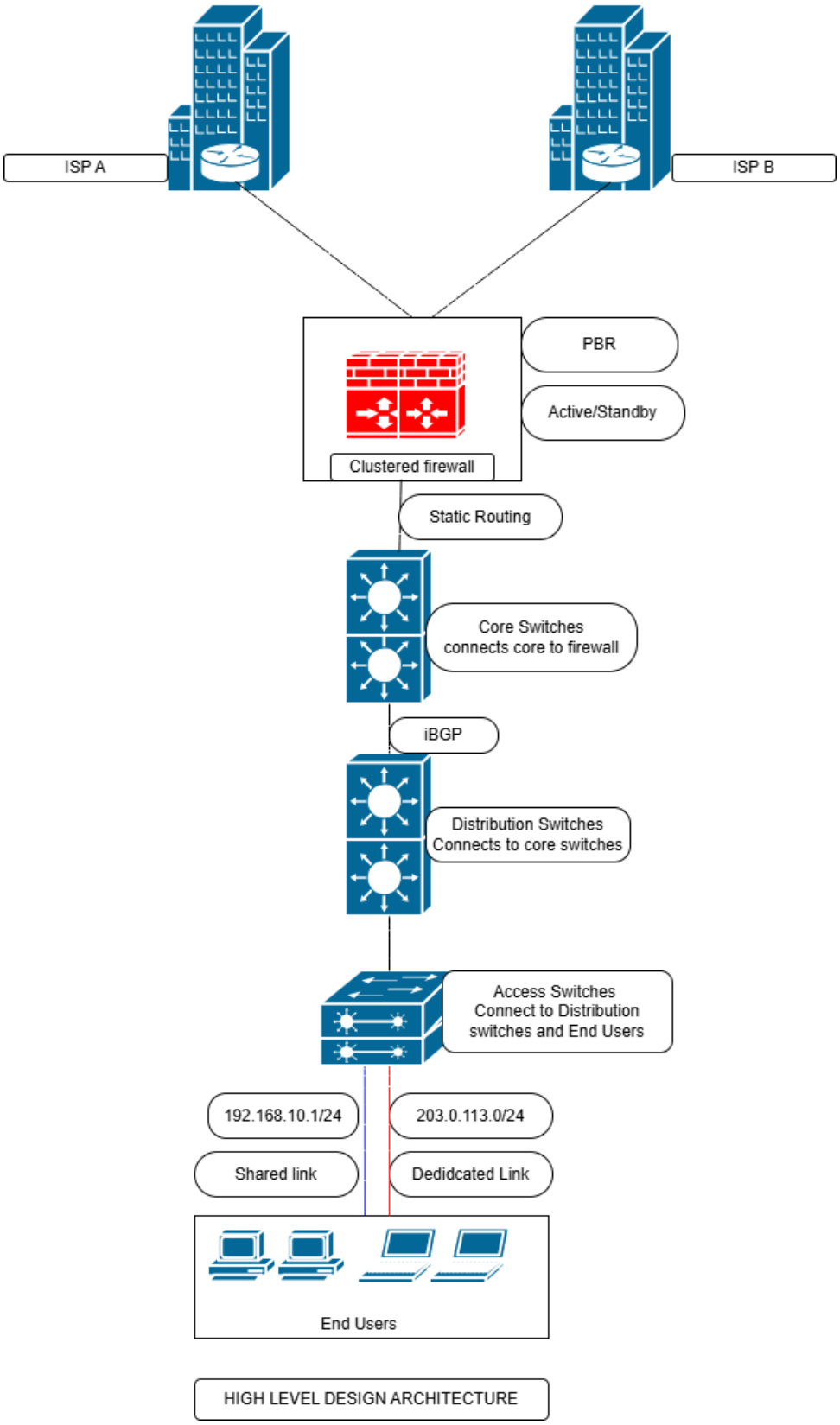


Figure 3: Network Topology Overview

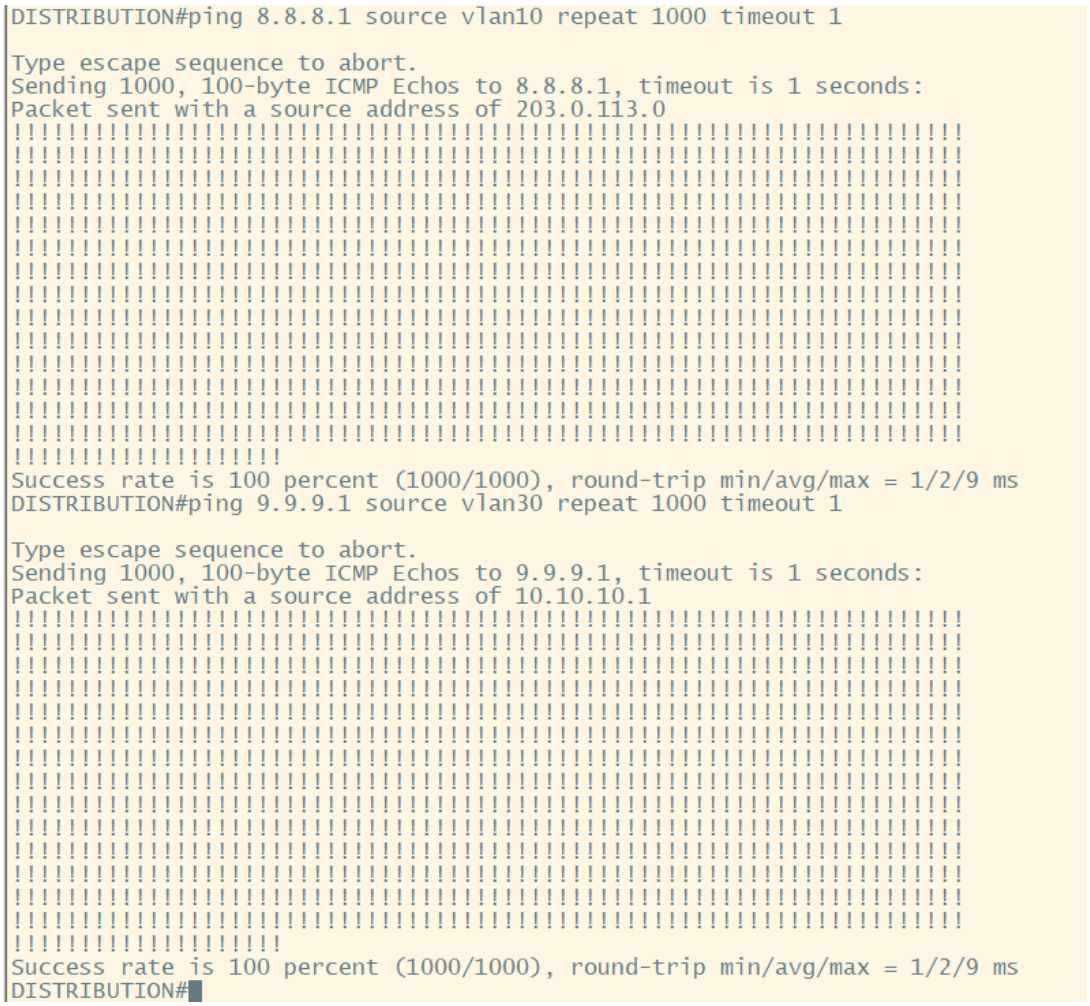


Figure 4: Normal operation: Dedicated and shared service traffic via ISP-RED and ISP-BLUE

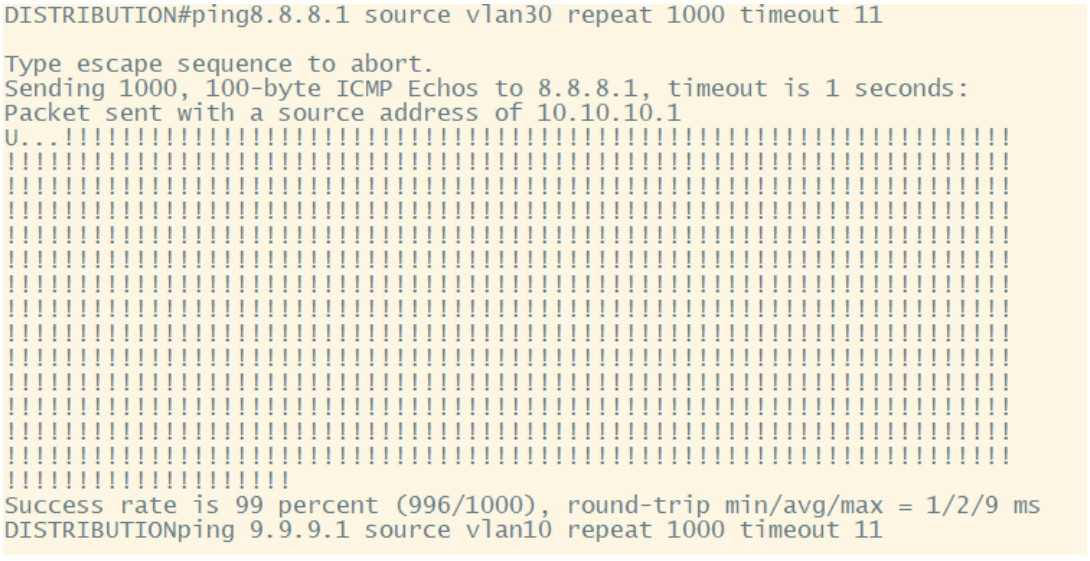


Figure 5: Shared service failover to ISP-RED upon ISP-BLUE failure

```
DISTRIBUTION#ping 9.9.9.1 source vlan10 repeat 1000 timeout 1

Type escape sequence to abort.
Sending 1000, 100-byte ICMP Echos to 9.9.9.1, timeout is 1 seconds:
Packet sent with a source address of 203.0.113.0
U...!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
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!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
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!!!!!!!!!!!!!!!!!!!!
Success rate is 99 percent (996/1000), round-trip min/avg/max = 1/2/26 ms
DISTRIBUTION#
```

Figure 6: Dedicated service failover to ISP-BLUE upon ISP-RED failure

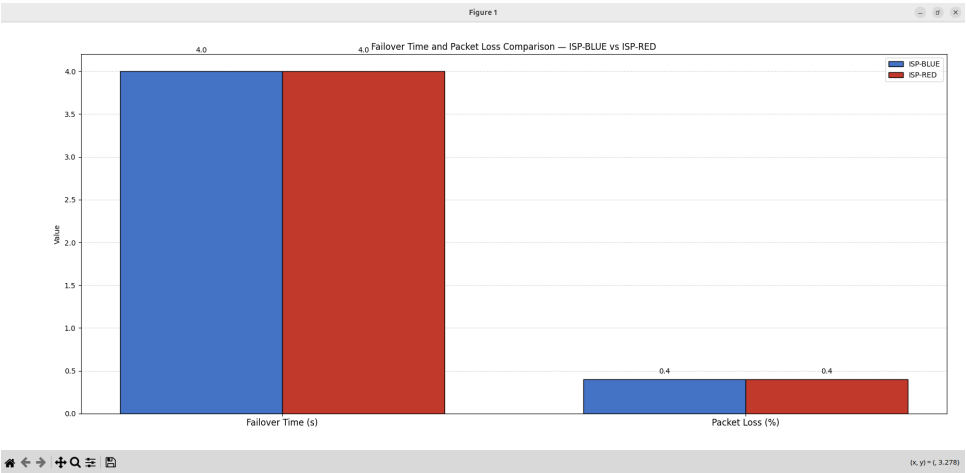


Figure 7: Packet loss observed during ISP failover scenarios

```
IPSLA operation id: 10
Type of operation: udp-jitter
        Latest RTT: 1 milliseconds
Latest operation start time: 18:31:07 UTC Wed May 13 2026
Latest operation return code: OK
RTT Values:
        Number Of RTT: 100              RTT Min/Avg/Max: 1/1/2 milliseconds
Latency one-way time:
        Number of Latency one-way Samples: 0
        Source to Destination Latency one way Min/Avg/Max: 0/0/0 milliseconds
        Destination to Source Latency one way Min/Avg/Max: 0/0/0 milliseconds
Jitter Time:
        Number of SD Jitter Samples: 99
        Number of DS Jitter Samples: 99
        Source to Destination Jitter Min/Avg/Max: 0/0/0 milliseconds
        Destination to Source Jitter Min/Avg/Max: 0/1/1 milliseconds
Over Threshold:
        Number Of RTT Over Threshold: 0 (0%)
Packet Loss Values:
        Loss Source to Destination: 0
        Source to Destination Loss Periods Number: 0
        Source to Destination Loss Period Length Min/Max: 0/0
        Source to Destination Inter Loss Period Length Min/Max: 0/0
        Loss Destination to Source: 0
        Destination to Source Loss Periods Number: 0
        Destination to Source Loss Period Length Min/Max: 0/0
        Destination to Source Inter Loss Period Length Min/Max: 0/0
        Out Of Sequence: 0      Tail Drop: 0
        Packet Late Arrival: 0  Packet Skipped: 0
Voice Score Values:
        Calculated Planning Impairment Factor (ICPIF): 0
        Mean Opinion Score (MOS): 0
Number of successes: 4
Number of failures: 13
Operation time to live: Forever
```

Figure 8: IP SLA UDP jitter statistics output from EDGE-FW

```
EDGE-FW#show ip sla statistics 11
*May 13 20:15:41.681: %TRACK-6-STATE: 1 ip sla 1 reachability Down -> Up
IPSLAs Latest Operation Statistics

IPSLA operation id: 11
Type of operation: udp-jitter
        Latest RTT: 1 milliseconds
Latest operation start time: 20:15:40 UTC Wed May 13 2026
Latest operation return code: OK
RTT Values:
        Number Of RTT: 100              RTT Min/Avg/Max: 1/1/1 milliseconds
Latency one-way time:
        Number of Latency one-way Samples: 0
        Source to Destination Latency one way Min/Avg/Max: 0/0/0 milliseconds
        Destination to Source Latency one way Min/Avg/Max: 0/0/0 milliseconds
Jitter Time:
        Number of SD Jitter Samples: 99
        Number of DS Jitter Samples: 99
        Source to Destination Jitter Min/Avg/Max: 0/0/0 milliseconds
        Destination to Source Jitter Min/Avg/Max: 0/0/0 milliseconds
Over Threshold:
        Number Of RTT Over Threshold: 0 (0%)
Packet Loss Values:
        Loss Source to Destination: 0
        Source to Destination Loss Periods Number: 0
        Source to Destination Loss Period Length Min/Max: 0/0
        Source to Destination Inter Loss Period Length Min/Max: 0/0
        Loss Destination to Source: 0
        Destination to Source Loss Periods Number: 0
        Destination to Source Loss Period Length Min/Max: 0/0
        Destination to Source Inter Loss Period Length Min/Max: 0/0
        Out Of Sequence: 0      Tail Drop: 0
        Packet Late Arrival: 0  Packet Skipped: 0
Voice Score Values:
        Calculated Planning Impairment Factor (ICPIF): 0
        Mean Opinion Score (MOS): 0
Number of successes: 1
Number of failures: 3
Operation time to live: Forever
```

Figure 9: IP SLA UDP jitter statistics output from EDGE-FW